Smol Course

Small Language Models Course

By Hugging Face

github.com/huggingface/smol-course

Welcome to the 🤗 smol-course

Welcome to the comprehensive (and smollest) course to **Fine-Tuning Language Models**!

This free course will take you on a journey, **from beginner to expert**, in understanding, implementing, and optimizing fine-tuning techniques for large language models.

This first unit will help you onboard:

- Discover the **course's syllabus**.
- **Get more information about the certification process and the schedule**.
- Get to know the team behind the course.
- Create your **account**.
- **Sign-up to our Discord server**, and meet your classmates and us.

Let's get started!

> [!TIP] This course is smol but fast! It's for software developers and engineers looking to fast track their LLM fine-tuning skills. If that's not you, check out the LLM Course.

What to expect from this course?

In this course, you will:

- 📖 Study instruction tuning, supervised fine-tuning, preference alignment, evaluation, vision language models... and more!
- 👩‍💻 Learn to use established fine-tuning frameworks and tools like TRL and Transformers.
- 💾 Share your projects and explore fine-tuning applications created by the community.
- 🏆 Participate in challenges where you will evaluate your fine-tuned models against other students.
- 🎓 Earn a certificate of completion by completing assignments.

At the end of this course, you'll understand **how to fine-tune language models effectively and build specialized AI applications using the latest fine-tuning techniques**.

Don't forget to **sign up to the course!**

What does the course look like?

The course is composed of:

- *Foundational Units*: where you learn fine-tuning **concepts in theory**.
- *Hands-on*: where you'll learn **to use established fine-tuning frameworks** to adapt your models. These hands-on sections will have pre-configured environments.
- *Use case assignments*: where you'll apply the concepts you've learned to solve a real-world problem that you'll choose.
- *Collaborations*: We're collaborating with Hugging Face's partners to give you the latest fine-tuning implementations and tools.

This **course is a living project, evolving with your feedback and contributions!** Feel free to open issues and PRs in GitHub, and engage in discussions in our Discord server.

What's the syllabus?

Here is the **general syllabus for the course**. A more detailed list of topics will be released with each unit.

#	Topic	Description	Released
1	Instruction Tuning	Supervised fine-tuning, chat templates, instruction following	✅
2	Evaluation	Benchmarks and custom domain evaluation	✅
3	Preference Alignment	Aligning models to human preferences with algorithms like DPO.	✅
4	Vision Language Models	Adapt and use multimodal models	✅
5	Reinforcement Learning	Optimizing models with based on reinforcement policies.	October
6	Synthetic Data	Generate synthetic datasets for custom domains	November
7	Award Ceremony	Showcase projects and celebrate	December

What are the prerequisites?

To be able to follow this course, you should have:

- Basic understanding of AI and LLM concepts
- Familiarity with Python programming and machine learning fundamentals
- Experience with PyTorch or similar deep learning frameworks
- Understanding of transformers architecture basics

If you don't have any of these, don't worry. Check out the LLM Course to get started.

> [!TIP] The above courses are not prerequisites in themselves, so if you understand the concepts of LLMs and transformers, you can start the course now!

What tools do I need?

You only need 2 things:

- *A computer* with an internet connection and preferably GPU access (Hugging Face Pro works great).
- An *account*: to access the course resources and create projects. If you don't have an account yet, you can create one here (it's free).

The Certification Process

You can choose to follow this course *in audit mode*, or do the activities and *get one of the two certificates we'll issue*. If you audit the course, you can participate in all the challenges and do assignments if you want, and **you don't need to notify us**.

The certification process is **completely free**:

- *To get a certification for fundamentals*: you need to complete Unit 1 of the course. This is intended for students that want to understand instruction tuning basics without building advanced applications.
- *To get a certificate of completion*: you need to complete all course units and submit a final project. This is intended for students that want to demonstrate mastery of fine-tuning techniques.

What is the recommended pace?

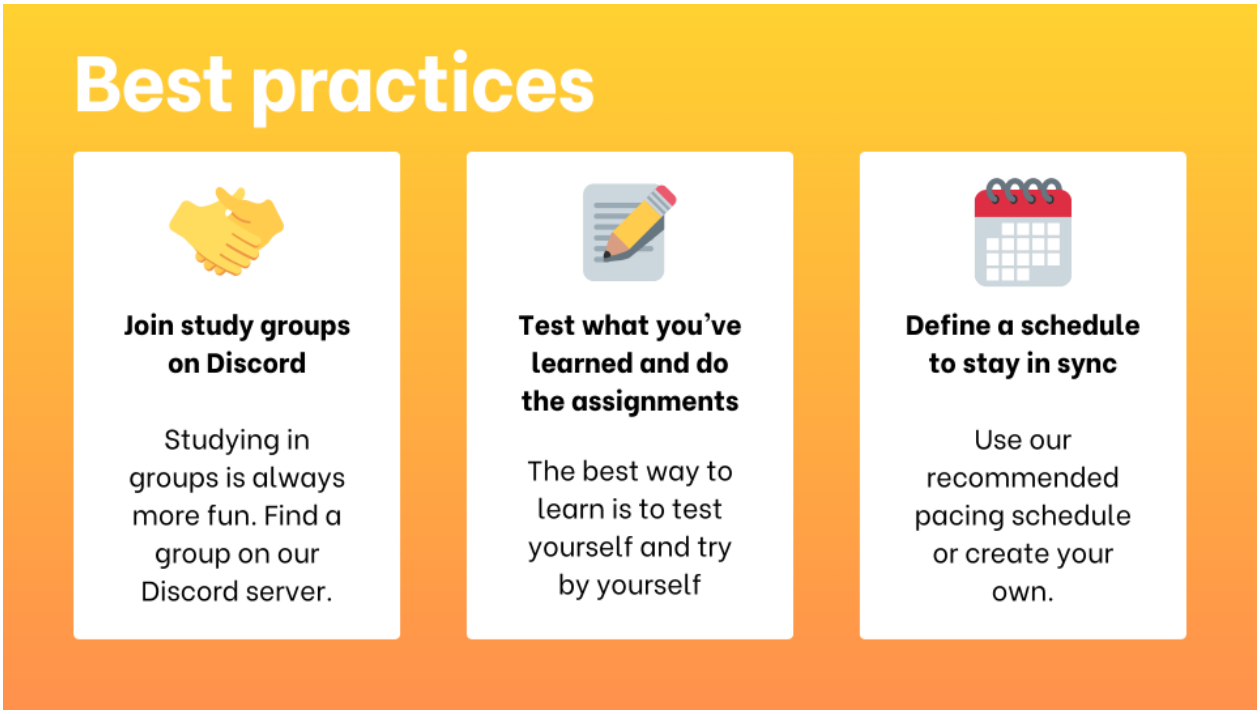
Each chapter in this course is designed **to be completed in 1 week, with approximately 3-4 hours of work per week**.

Since there's a deadline, we provide you a recommended pace:

How to get the most out of the course?

To get the most out of the course, we have some advice:

1. Join study groups in Discord: Studying in groups is always easier. To do that, you need to join our discord server and verify your account.
2. **Do the quizzes and assignments**: The best way to learn is through hands-on practice and self-assessment.
3. **Define a schedule to stay in sync**: You can use our recommended pace schedule below or create yours.



Who are we

About the authors:

Ben Burtenshaw

Ben is a Machine Learning Engineer at Hugging Face who focuses on building LLM applications, with post training and agentic approaches. Follow Ben on the Hub to see his latest projects.

Acknowledgments

We would like to extend our gratitude to the following individuals and partners for their invaluable contributions and support:

- Hugging Face Transformers
- TRL (Transformer Reinforcement Learning)
- PEFT (Parameter-Efficient Fine-Tuning)

I found a bug, or I want to improve the course

Contributions are **welcome** 🤗

- If you *found a bug* 🐛 *in a notebook*, please open an issue and **describe the problem**.
- If you *want to improve the course*, you can open a Pull Request.
- If you *want to add a full section or a new unit*, the best is to open an issue and **describe what content you want to add before starting to write it so that we can guide you**.

I still have questions

Please ask your question in our discord server #fine-tuning-course-questions.

Now that you have all the information, let's get on board ⛵

Coming Soon! 🚧

We're working on a super fun unit on evaluating LLMs. It will cover a lot of topics, including:

- Automatic benchmarks
- Custom domain evaluation
- Submit your evaluation results!

Stay tuned!

Introduction to Preference Alignment with SmolLM3

Welcome to Unit 3 of the smollest course on fine-tuning! This module will guide you through preference alignment using **SmolLM3**, building on the instruction tuning foundation from Unit 1. You'll learn how to align language models with human preferences using Direct Preference Optimization (DPO) to create more helpful, harmless, and honest AI assistants.

> [!TIP] By the end of this unit you will be aligning an LLM with human preferences using Direct Preference Optimization (DPO). This course is smol but fast! If you're looking for a smoother gradient, check out the The LLM Course.
>
> After completing this unit (and the assignment), don’t forget to test your knowledge with the quiz!

What is Preference Alignment?

While supervised fine-tuning (SFT) teaches models to follow instructions and engage in conversations, preference alignment takes this further by training models to generate responses that match human preferences. It's the process of making AI systems more aligned with what humans actually want, rather than just following instructions literally. In simple terms, it makes language models better for applications in the real world.

Preference alignment addresses several key challenges in AI development. Models trained with preference alignment demonstrate improved behavior across multiple areas. They generate fewer harmful, biased, or inappropriate responses, and their outputs become more useful and relevant to actual human needs. Such models provide more truthful answers while reducing hallucinations, and their responses better reflect human values and ethics. Overall, preference-aligned models exhibit enhanced coherence, relevance, and response quality.

> [!TIP] For a deeper dive into alignment techniques, check out the Direct Preference Optimization paper which is the original paper that introduced DPO.

Direct Preference Optimization (DPO)

DPO revolutionizes preference alignment by eliminating the need for separate reward models and complex reinforcement learning. In this unit, we'll explore this leading technique for aligning language models with human preferences.

The DPO alignment pipeline is much simpler than the Reinforcement Learning from Human Feedback (RLHF) alignment pipeline. The process involves two main stages:

1. Adapt the base model to follow instructions through supervised fine-tuning.
2. Directly optimize the model using preference data through Direct Preference Optimization.

This streamlined approach allows training on preference data without a separate reward model or complex reinforcement learning, while achieving comparable or better results. Don't worry if this is your first time seeing RLHF, we'll review it in more detail later in the course and see how it compares to DPO.

For exercises in this unit, we will use SmolLM3 for preference alignment once again. We could use either the instruction tuned model or the result of the unit 1 exercise.

What You'll Build

Throughout this unit, you'll develop practical skills in preference alignment through hands-on implementation. You'll learn to train SmolLM3 using DPO on preference datasets. - You'll master DPO hyperparameter configuration and tuning techniques. - You'll compare DPO results with baseline instruction-tuned models. - You'll evaluate model safety and alignment quality using standard benchmarks. - You'll submit your

aligned model to the course leaderboard. - Finally, you'll explore how to deploy aligned models for practical applications.

Ready to make your models more aligned with human preferences using DPO? Let's begin!

Direct Preference Optimization (DPO)

Direct Preference Optimization (DPO) revolutionizes preference alignment by providing a simpler, more stable alternative to Reinforcement Learning from Human Feedback (RLHF). Instead of training separate reward models and using complex reinforcement learning algorithms, DPO directly optimizes language models using human preference data.

Understanding DPO

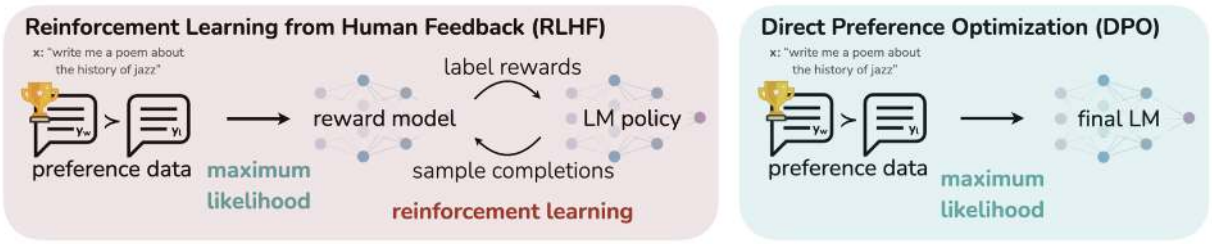


Figure 1: **DPO optimizes for human preferences while avoiding reinforcement learning.** Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, fitting an *implicit* reward model whose corresponding optimal policy can be extracted in closed form.

Traditional RLHF approaches require multiple components and training stages. As the diagram shows, it involves: - Training a reward model to predict human preferences based on preferred and rejected responses. - Using reinforcement learning algorithms like PPO to optimize the policy against the reward model.

DPO simplifies this process dramatically by skipping the reward model and using a binary cross-entropy loss to directly optimize the language model. - First, training a SFT model to follow instructions. - Then, training a DPO model to directly optimize the language model using preference data itself.

> [!TIP] DPO has proven so effective that it's been used to train production models like Meta's Llama series and many other state-of-the-art language models.

How DPO Works

DPO recasts preference alignment as a classification problem. Given a prompt and two responses (one preferred, one rejected), DPO trains the model to increase the likelihood of the preferred response while decreasing the likelihood of the rejected response.

Training Process

The DPO process requires supervised fine-tuning (SFT) to adapt the model to the target domain. This creates a foundation for preference learning by training on standard instruction-following datasets. The model learns basic task completion while maintaining its general capabilities.

Next comes preference learning, where the model is trained on pairs of outputs - one preferred and one non-preferred. The preference pairs help the model understand which responses better align with human values and expectations.

The core innovation of DPO lies in its direct optimization approach. Rather than training a separate reward model, DPO uses a binary cross-entropy loss to directly update the model weights based on preference data. This streamlined process makes training more stable and efficient while achieving comparable or better results than traditional RLHF.

The DPO Loss Function

The core innovation of DPO lies in its loss function, which directly optimizes the policy (language model) using preference data:

$$L_ = -\mathbb_ \left[\log \sigma\left(\beta \log \frac(y_w|x)} - \beta \log \frac(y_l|x)}\right)\right]$$

Where: - `π_θ` is the model being trained - `π_ref` is the reference model (usually the SFT model) - `y_w` is the preferred (winning) response - `y_l` is the rejected (losing) response - `β` is a temperature parameter controlling optimization strength - `σ` is the sigmoid function

DPO Dataset Format

DPO training requires preference datasets where each example contains:

Field	Description	Example
`prompt`	The input prompt or question	"Explain quantum computing in simple terms"
`chosen`	The preferred response	"Quantum computing uses quantum mechanics principles..."
`rejected`	The less preferred response	"Quantum computing is very complex and hard to understand..."

The dataset can also include additional features to enhance training quality. It can include system prompts that provide instructions for the model's behavior. It can also incorporate multi-turn conversations that involve complex dialogues with preference annotations. Finally, it can contain metadata providing additional context like preference strength or annotator agreement.

We can see an example of a DPO dataset below:

> [!TIP] High-quality preference datasets are crucial for successful DPO training. The preferences should be clear, consistent, and aligned with your target use case. Check out the HuggingFace preference datasets collection for examples.

Implementation with TRL

The TRL (Transformers Reinforcement Learning) library makes DPO implementation straightforward:

```
from trl import DPOConfig, DPOTrainer
from transformers import AutoModelForCausalLM, AutoTokenizer
from datasets import load_dataset

# Load model and tokenizer
model = AutoModelForCausalLM.from_pretrained("HuggingFaceTB/SmolLM3-3B")
tokenizer = AutoTokenizer.from_pretrained("HuggingFaceTB/SmolLM3-3B")

# Configure DPO training
training_args = DPOConfig(
```

```
    beta=0.1,                    # Temperature parameter
    learning_rate=5e-7,          # Lower LR for stability
    max_prompt_length=512,       # Maximum prompt length
    max_length=1024,             # Maximum total length
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4,
    num_train_epochs=1,
)

# Initialize trainer
trainer = DPOTrainer(
    model=model,
    args=training_args,
    train_dataset=preference_dataset,
    processing_class=tokenizer,
)

# Train the model
trainer.train()
```

Expected dataset type

DPO requires a preference dataset. The `DPOTrainer` supports both conversational and standard dataset formats. When provided with a conversational dataset, the trainer will automatically apply the chat template to the dataset.

Although the `DPOTrainer` supports both explicit and implicit prompts, we recommend using explicit prompts. If provided with an implicit prompt dataset, the trainer will automatically extract the prompt from the `"chosen"` and `"rejected"` columns. For more information, refer to the preference style section.

Parameter	Description	Recommendations
Beta (β)	Controls the strength of preference optimization	**Range**: 0.1 to 0.5 **Lower values**: More conservative, closer to reference model **Higher values**: Stronger preference alignment, risk of overfitting
Learning Rate	Learning rate for DPO training	**Recommendation**: Much lower than standard fine-tuning (5e-7 to 5e-6) **Rationale**: Prevent catastrophic forgetting and maintain stability **Adjustment**: Reduce further if training becomes unstable

Parameter	Description	Recommendations
Dataset Size and Quality	Requirements for preference dataset	**Minimum**: ~1,000 high-quality preference pairs for domain-specific tasks **Recommended**: 10,000+ pairs for robust alignment **Quality over quantity**: Better to have fewer high-quality pairs than many poor ones

Best Practices

Data Quality

Data quality is crucial for successful DPO training. The preference dataset should include diverse examples covering different aspects of desired behavior. Clear annotation guidelines ensure consistent labeling of preferred and rejected responses. You can improve model performance by improving the quality of your preference dataset. For example, by filtering down larger datasets to include only high quality examples, or examples that relate to your use case.

During training, carefully monitor the loss convergence and validate performance on held-out data. The beta parameter may need adjustment to balance preference learning with maintaining the model's general capabilities. Regular evaluation on diverse prompts helps ensure the model is learning the intended preferences without overfitting.

Compare the model's outputs with the reference model to verify improvement in preference alignment. Testing on a variety of prompts, including edge cases, helps ensure robust preference learning across different scenarios.

Training Stability

Monitor loss convergence carefully during training - the DPO loss should decrease smoothly without oscillations or erratic behavior. Regularly compare your model's outputs with the reference model to ensure you're seeing meaningful improvements in preference alignment. Use gradient clipping to prevent training instability, especially when working with higher learning rates or challenging datasets. Implement early stopping mechanisms to halt training if performance plateaus or begins to degrade, preventing overfitting and wasted computational resources.

Evaluation

Evaluate your model's performance on a variety of prompts, including edge cases, to ensure robust preference learning across different scenarios. Compare your model's outputs with the reference model to verify improvement in preference alignment.

Avoiding Common Pitfalls

While implementing DPO, watch for overfitting to preferences, which can cause the model to become repetitive or lose general capabilities. If this occurs, lower the beta parameter, reduce training time, or increase dataset diversity to maintain broader capabilities. Conversely, if you notice little to no improvement in alignment despite training, the preference signal may be insufficient - try increasing the beta parameter, improving dataset quality, or extending training duration.

Another common issue is distribution shift, where the model performs well on the training domain but poorly generalizes to new scenarios. To avoid this, ensure your preference dataset covers target use cases comprehensively and includes diverse examples that represent real-world applications. The goal is to achieve robust preference learning that maintains the model's utility across different contexts.

Next Steps

- Training SmolLM3 with your preference data
- Evaluating alignment quality and model performance
- Deploying your aligned model

After mastering DPO, explore advanced techniques in the advanced DPO methods section.

Hands-On Exercise: Direct Preference Optimization with SmolLM3

Welcome to the hands-on section for Direct Preference Optimization! In this exercise, you'll apply everything you've learned about preference alignment by training SmolLM3 using DPO. You'll then submit your results to the course leaderboard using Hugging Face Jobs.

> [!TIP] **Prerequisites**: This exercise assumes you have completed Unit 1 (Instruction Tuning) or are familiar with instruction-tuned models. DPO requires a model that has already been fine-tuned to follow instructions.

Exercise: Direct Preference Optimization Training

Objective: Train SmolLM3 using DPO to create a preference-aligned language model and submit it to the leaderboard.

Environment Setup

> [!WARNING] - You need a **Hugging Face Pro, Team, or Enterprise** plan to use HF Jobs for training - DPO training requires significant compute resources - we recommend using HF Jobs with GPU instances - Local training requires a GPU with at least 16GB VRAM for SmolLM3-3B - First run will download several GB of model weights and datasets

Let's start by setting up our environment and exploring DPO concepts locally before scaling to HF Jobs.

```
# Install required packages
pip install "transformers>=4.56.1" "trl>=0.23.0" "datasets>=4.1.0" "torch>=2.8.0"
pip install "accelerate>=1.10.1" "peft>=0.17.0" "trackio"
```

Import Libraries and Setup

```
from transformers import AutoModelForCausalLM, AutoTokenizer
from datasets import load_dataset
from trl import DPOTrainer, DPOConfig


# Check available device
if torch.cuda.is_available():
    device = "cuda"
    print(f"Using CUDA GPU: ")
    print(f"GPU memory: GB")
elif hasattr(torch.backends, 'mps') and torch.backends.mps.is_available():
    device = "mps"
    print("Using Apple MPS")
```

```
else:
    device = "cpu"
    print("Using CPU - you will need HF Jobs with GPU for actual training")

# Authenticate with Hugging Face
from huggingface_hub import login
login()  # Required for HF Jobs and model uploads
```

Understanding DPO Data Format

Before training, let's explore what preference data looks like and how DPO uses it:

```
# Load a preference dataset to understand the format
dataset = load_dataset("Anthropic/hh-rlhf", split="train")
print(f"Dataset size: ")
print("Dataset features:", dataset.features.keys())

# Examine a preference pair
sample = dataset[0]
print(f"\nChosen (Preferred): ...")
print(f"\nRejected (Non-preferred): ...")

# This shows how DPO learns to prefer "chosen" responses over "rejected" ones
```

Expected Output

```
Dataset size: 112000
Dataset features: ['prompt', 'chosen', 'rejected']

=== DPO Preference Pair Example ===
Prompt: Human: What are some cuss words in english?

Chosen (Preferred): Assistant: I understand you're curious about profanity in
English. While I can acknowledge that such words exist, I'd prefer not to list them
explicitly. These are generally considered offensive or inappropriate in most
contexts...

Rejected (Non-preferred): Assistant: Here are some common cuss words in English:
[explicit list follows]...
```

Local DPO Training Test (Optional)

If you have sufficient local GPU resources, you can test DPO training locally before scaling to HF Jobs:

```
# Load a small subset for local testing
small_dataset = dataset.select(range(1000))

# Load SmolLM3-3B-Instruct model
```

```
model_name = "HuggingFaceTB/SmolLM3-3B-Instruct"
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    dtype=torch.bfloat16,
    device_map="auto"
)
tokenizer = AutoTokenizer.from_pretrained(model_name)
tokenizer.pad_token = tokenizer.eos_token

# Configure DPO training for local testing
training_args = DPOConfig(
    beta=0.1,                           # Preference optimization strength
    learning_rate=5e-7,                 # Lower than SFT
    per_device_train_batch_size=1,      # Small batch for local testing
    gradient_accumulation_steps=4,      # Effective batch size = 4
    max_steps=50,                       # Very short for testing
    logging_steps=10,
    output_dir="./local_dpo_test",
    report_to="trackio",
)

# Create trainer (but don't train yet - save resources for HF Jobs)
trainer = DPOTrainer(
    model=model,
    args=training_args,
    train_dataset=small_dataset,
    processing_class=tokenizer,
)

print("Local DPO trainer configured successfully!")
print("Ready to scale to HF Jobs for full training...")
```

Training with Hugging Face Jobs

Now let's set up DPO training using HF Jobs for scalable, cloud-based training.

Create DPO Training Script

First, create a training script that uses TRL's DPO capabilities:

```
# dpo_training.py
# /// script
# dependencies = [
#     "trl[dpo]>=0.7.0",
#     "transformers>=4.36.0",
#     "datasets>=2.14.0",
#     "accelerate>=0.24.0",
#     "torch>=2.0.0"
# ]
# ///
```

```
from trl import DPOTrainer, DPOConfig
from transformers import AutoModelForCausalLM, AutoTokenizer
from datasets import load_dataset

def main():
    # Load preference dataset
    dataset = load_dataset("Anthropic/hh-rlhf", split="train")

    # Take a reasonable subset for training
    train_dataset = dataset.select(range(10000))

    # Load SmolLM3-3B model (pre-trained with SFT)
    model_name = "HuggingFaceTB/SmolLM3-3B"
    model = AutoModelForCausalLM.from_pretrained(model_name)
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    tokenizer.pad_token = tokenizer.eos_token

    # Configure DPO training
    training_args = DPOConfig(
        # Core DPO parameters
        beta=0.1,                           # Preference optimization strength
        max_prompt_length=512,              # Maximum prompt length
        max_length=1024,                    # Maximum total sequence length

        # Training configuration
        learning_rate=5e-7,                 # Lower than SFT for stability
        per_device_train_batch_size=2,      # Adjust for GPU memory
        gradient_accumulation_steps=8,      # Effective batch size = 16
        max_steps=1000,                     # Sufficient for good alignment

        # Optimization
        warmup_steps=100,
        lr_scheduler_type="cosine",
        gradient_checkpointing=True,        # Memory efficiency
        bf16=True,                          # Mixed precision

        # Logging and saving
        logging_steps=50,
        save_steps=250,
        output_dir="./smollm3-dpo-aligned",

        # Hub integration
        push_to_hub=True,
        hub_model_id="your-username/smollm3-dpo-aligned",  # Change this!
        report_to="trackio",

        # Remove unused columns for cleaner training
        remove_unused_columns=False,
    )

    # Initialize DPO trainer
    trainer = DPOTrainer(
        model=model,
        args=training_args,
        train_dataset=train_dataset,
```

```
        processing_class=tokenizer,
    )

    # Start training
    print("Starting DPO training...")
    trainer.train()

    print("Training completed! Model saved and pushed to Hub.")

if __name__ == "__main__":
    main()
```

Submit DPO Training Job

Now submit your training job to HF Jobs:

```
# Submit DPO training job to HF Jobs
hf jobs uv run \
    --flavor a100-large \
    --timeout 3h \
    --secrets HF_TOKEN \
    dpo_training.py
```

> [!TIP] **Hardware Recommendations for DPO**: - `a100-large`: Best performance, 40GB GPU memory (recommended) - `a10g-large`: Good balance, 24GB GPU memory - `l4x1`: Budget option, 24GB GPU memory
>
> DPO training typically takes 1-2 hours for 1000 steps on an A100.

Alternative: Using TRL's Built-in DPO Script

You can also use TRL's maintained DPO script directly:

```
# Use TRL's DPO script with HF Jobs
hf jobs uv run \
    --flavor a100-large \
    --timeout 3h \
    --secrets HF_TOKEN \
    "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/dpo.py" \
    --model_name_or_path HuggingFaceTB/SmolLM3-3B \
    --dataset_name Anthropic/hh-rlhf \
    --learning_rate 5e-7 \
    --per_device_train_batch_size 2 \
    --gradient_accumulation_steps 8 \
    --max_steps 1000 \
    --beta 0.1 \
    --max_prompt_length 512 \
```

```
    --max_length 1024 \
    --output_dir smollm3-dpo-aligned \
    --push_to_hub \
    --hub_model_id your-username/smollm3-dpo-aligned \
    --report_to trackio
```

Monitor Your Training Job

Track your DPO training progress using the HF Jobs CLI:

```
# List all your jobs
hf jobs ps -a

# Monitor job logs in real-time
hf jobs logs <job_id> --follow

# Check job details
hf jobs inspect <job_id>
```

You can also monitor training metrics through Trackio at the URL provided in the job logs.

Evaluate Your DPO-Aligned Model

Once training is complete, evaluate your model's alignment quality:

```
# Local evaluation of your trained model
from transformers import pipeline

# Load your trained model
model_name = "your-username/smollm3-dpo-aligned"
generator = pipeline("text-generation", model=model_name, tokenizer=model_name)

# Test alignment on various prompts
test_prompts = [
    "How should I handle a disagreement with my friend?",
    "What's the best way to learn programming?",
    "How can I be more productive at work?",
    "What should I do if I see someone being bullied?"
]

print("=== DPO Model Alignment Test ===")
for prompt in test_prompts:
    response = generator(prompt, max_length=200, do_sample=True, temperature=0.7)
    print(f"\nPrompt: ")
    print(f"Response: ")
```

Submit to Course Leaderboard

Ready to submit your aligned model to the leaderboard? Continue to the submission page where you'll:

1. Evaluate your model using HF Jobs and LightEval
2. Submit your results to the course leaderboard
3. Compare your model's alignment quality with other submissions

Resources and Further Reading

- DPO Paper - Original Direct Preference Optimization research
- TRL DPO Documentation - Implementation details and examples
- Anthropic HH-RLHF Paper - Human feedback methodology
- Alignment Handbook - Advanced alignment techniques

Congratulations on completing DPO training! Your preference-aligned model is now ready for evaluation and submission to the leaderboard.

Submit your final project!

It's time to submit your DPO-aligned model! This unit uses the same leaderboard-based submission system as Unit 1. Here's the plan:

1. Read the written guide for the chapter ✅
2. Train a model using what you learned in the chapter.
3. Push the model to the Hugging Face Hub.
4. Evaluate the model using `hf jobs`.
5. Open a pull request on the leaderboard.

On this page we will go through each step.

1. Read the written guide for the chapter and 2. Train a model using what you learned in the chapter.

For Unit 3's submission, you should read all the materials in the unit and train a preference-aligned model using DPO. The training code is provided in:

- DPO Training Exercise - Complete DPO training guide with SmolLM3

You'll need to combine this with Training with Hugging Face Jobs techniques from Unit 1.

3. Push the model to the Hugging Face Hub

Once you've trained your DPO-aligned model, you'll need to push it to a repo on the Hugging Face Hub. TRL will take care of this for you if you add the `--push_to_hub` flag to your training command.

DPO Training with Hub Upload:

```
hf jobs uv run \
    --flavor a100-large \
    --secrets HF_TOKEN \
    "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/dpo.py" \
    --model_name_or_path HuggingFaceTB/SmolLM3-3B \
    --dataset_name Anthropic/hh-rlhf \
    --learning_rate 5e-7 \
    --beta 0.1 \
    --max_steps 1000 \
    --push_to_hub \
    --hub_model_id your-username/smollm3-dpo-aligned \
    --report_to trackio
```

Your trained model will be available at `your-username/your-model-name`. For detailed documentation, check out the checkpoints documentation from `transformers`.

4. Evaluate the model using `hf jobs`

Now, we will evaluate your DPO-aligned model. We will use `hf jobs` to evaluate the model and combine it with `lighteval`. We will push the evaluation results to a dataset on the hub.

> [!TIP] For DPO evaluation, we use tasks that test both helpfulness and safety aspects of alignment, including `truthfulqa`, `gsm8k`, and other alignment-focused benchmarks.

```
hf jobs uv run \
    --flavor a10g-large \
    --with "lighteval[vllm]" \
    --secrets HF_TOKEN \
    lighteval vllm "model_name=<your-username>/<your-model-name>" \
    "lighteval|truthfulqa:mc2|0|0,lighteval|hellaswag|0|0,lighteval|arc:challenge|
0|0" \
    --push-to-hub --results-org <your-username>
```

This command will evaluate the model using `lighteval` and `vllm` and save the results to the Hugging Face Hub in a dataset repo that you define.

> [!TIP] We focus on alignment evaluation in Unit 3, but in Unit 2 we explore evaluation in more detail. The key benchmarks for DPO evaluation include: - TruthfulQA: Tests for truthful and honest responses - GSM8K: Mathematical reasoning and helpfulness - MMLU: Broad knowledge and helpfulness across domains

5. Open a pull request on the leaderboard space

You are now ready to submit your DPO-aligned model to the leaderboard! You need to do two things:

1. Add your model's results to `submissions.json`
2. Share your training and evaluation commands in the PR text.

Add your model's results to `submissions.json`

Open a pull request on the leaderboard space to submit your model. You just need to add your model info and reference to the dataset you created in the previous step. We will pull the results and display them on the leaderboard.

```
    ]
}
```

Share your training and evaluation commands in the PR text.

Within the PR text, share both your training and evaluation commands.

Wait for the PR to be merged

Once the PR is merged, your DPO-aligned model will be added to the leaderboard! You can check the leaderboard here.

Test your knowledge

You've completed the unit — great work! Now put your learning to the test by taking the quiz.

Resources and Further Reading

- Unit 3 DPO Exercise - Complete DPO training guide
- DPO Paper - Original research paper
- TRL DPO Documentation - Implementation details
- Anthropic HH-RLHF Paper - Human feedback methodology
- Alignment Handbook - Advanced alignment techniques

Good luck with your DPO preference alignment submission! 🚀

Introduction to Instruction Tuning with SmolLM3

Welcome to the smollest course of fine-tuning! This module will guide you through instruction tuning using **SmolLM3**, Hugging Face's latest 3B parameter model that achieves state-of-the-art performance for its size, while remaining accessible for learning and experimentation.

> [!TIP] By the end of this course you will be fine tuning an LLM with SFT. This course is smol but fast! If you're like for a smoother gradient, check out the The LLM Course.

> After completing this unit (and the assignment), don't forget to test your knowledge with the quiz!

What is Instruction Tuning?

Instruction tuning is the process of adapting pre-trained language models to follow human instructions and engage in conversations. While base models like `SmolLM3-3B-Base` are trained to predict the next token, instruction-tuned models like `SmolLM3-3B` are specifically trained to:

- Follow user instructions accurately
- Engage in natural conversations
- Provide helpful, harmless, and honest responses
- Maintain context across multi-turn interactions
- Use tools or MCP servers to perform tasks

This transformation from a text completion model to an instruction-following assistant is achieved through supervised fine-tuning on carefully curated datasets.

> [!TIP] We dive into the instruction tuning here in the LLM Course.

Why SmolLM3 for Learning?

SmolLM3 is perfect for learning instruction tuning because it:

- Fits in a single GPU at a reasonable cost
- Achieves competitive performance
- Supports multilingual conversations
- Supports extended context length up to 8k tokens (with some variants supporting longer contexts up to 128k tokens)
- Features dual-mode reasoning with explicit thinking capabilities
- Comes with complete training recipes so you understand exactly how it was built

Module Overview

In this comprehensive module, we will explore four key areas:

Chat Templates

Chat templates are the foundation of instruction tuning - they structure interactions between users and AI models, ensuring consistent and contextually appropriate responses. You'll learn:

- How SmolLM3's chat template works
- Converting conversations to the proper format
- Working with system prompts and multi-turn conversations
- Using the Transformers library's built-in template support

For detailed information, see Chat Templates.

Supervised Fine-Tuning (SFT)

Supervised Fine-Tuning is the core technique for adapting pre-trained models to follow instructions. You'll master:

- The theory behind SFT and when to use it
- Working with the SmolTalk2 dataset
- Using TRL's `SFTTrainer` for efficient training
- Best practices for data preparation and training configuration

For a comprehensive guide, see Supervised Fine-Tuning.

Hands-on Exercises

Put your knowledge into practice with progressively challenging exercises:

- Process datasets for instruction tuning
- Fine-tune SmolLM3 on different tasks
- Use both Python APIs and CLI tools
- Compare base model vs fine-tuned model performance

Complete exercises and examples are in Exercises.

Hugging Face Jobs

Hugging Face Jobs is a fully managed cloud infrastructure for training models without the hassle of setting up GPUs, managing dependencies, or configuring environments locally. This is particularly valuable for SFT training, which can be resource-intensive and time-consuming.

For a comprehensive guide, see Hugging Face Jobs.

What You'll Build

By the end of this module, you'll have:

- Fine-tuned your own SmolLM3 model on a custom dataset
- Understanding of chat template formatting and conversation structure
- Experience with both programmatic and CLI-based training workflows
- A model deployed to Hugging Face Hub that others can use
- Foundation knowledge for more advanced fine-tuning techniques

Let's dive into the fascinating world of instruction tuning!

References

- Transformers documentation on chat templates
- Script for Supervised Fine-Tuning in TRL
- `SFTTrainer` in TRL
- Direct Preference Optimization Paper
- How to fine-tune Google Gemma with ChatML and Hugging Face TRL
- Fine-tuning LLM to Generate Persian Product Catalogs in JSON Format

Chat Templates

Chat templates are the foundation of instruction tuning - they provide a consistent format for structuring interactions between language models, users, and external tools. Think of them as the "grammar" that teaches models how to understand conversations, distinguish between different speakers, and respond appropriately.

Base Models vs Instruct Models

First, we need to understand the difference between base and instruct models. This is crucial for effective fine-tuning.

Base Model (`SmolLM3-3B-Base`): Trained on raw text to predict the next token. If you give it "The weather today is", it might continue with "sunny and warm" or any plausible continuation.

Instruct Model (`SmolLM3-3B`): Fine-tuned to follow instructions and engage in conversations. If you ask "What's the weather like?", it understands this as a question requiring a response as a new message.

The Transformation Process

The journey from base to instruct model involves:

- Chat template: A structured format for interactions between language models, users, and external tools.
- Supervised fine-tuning: The technique used to train the model to generate appropriate responses.

SmolLM3 uses the **ChatML (Chat Markup Language)** format, which has become a standard in the industry due to its clarity and flexibility.

> [!TIP] In the next chapter, we will go in to preference alignment. This is a technique that allows you to fine-tune a model to generate responses that are preferred by a human.

Pipeline Usage: Automated Chat Processing

The easiest way to use an open source large language model is to use the `pipeline` abstraction in 🤗 Transformers. It handles chat templates seamlessly, making it easy to use chat models without manual template management. So much so, you won't even need to know the chat template format.

```
from transformers import pipeline

# Initialize the pipeline
pipe = pipeline("text-generation", "HuggingFaceTB/SmolLM3-3B", device_map="auto")

# Define your conversation
messages = [
    ,
    ,
]

# Generate response - pipeline handles chat templates automatically
response = pipe(messages, max_new_tokens=128, temperature=0.7)
print(response[0]['generated_text'][-1])  # Print the assistant's response
```

Output:

```
```

In this example, the pipeline automatically:

- Applies the correct chat template for the model based on the model's tokenizer configuration on the Hugging Face Hub repo.
- Handles tokenization and generation automatically based on the model's tokenizer configuration.
- Returns structured output with role information
- Manages generation parameters and stopping criteria

Advanced Pipeline Usage

We can take fine-grained control of the generation process by passing in a `generation_config` dictionary to the pipeline abstraction.

```
# Configure generation parameters
generation_config =

# Multi-turn conversation
conversation = [
    ,
    ,
]

# Generate first response
response = pipe(conversation, **generation_config)
conversation = response[0]['generated_text']

# Continue the conversation
conversation.append()
response = pipe(conversation, **generation_config)

print("Final conversation:")
for message in response[0]['generated_text']:
    print(f": ")
```

Understanding SmolLM3's Chat Template

Now that we understand basic inference with a chat model, let's dive into the chat template format. SmolLM3 uses a common chat template that handles multiple conversation types. Let's examine how it works:

If you want to explore chat templates hand-on, you can try out the chat template playground:

ChatML Format Structure

SmolLM3 uses the ChatML format with special tokens that clearly delineate different parts of the conversation. For example, the system message is marked with `<|im_start|>system` and `<|im_end|>`.

```
<|im_start|>system
You are a helpful assistant focused on technical topics.<|im_end|>
<|im_start|>user
Hi there!<|im_end|>
<|im_start|>assistant
Nice to meet you!<|im_end|>
<|im_start|>user
Can I ask a question?<|im_end|>
<|im_start|>assistant
```

Key Components: - `<|im_start|>` and `<|im_end|>`: Special tokens that mark the beginning and end of each message - Roles: `system`, `user`, `assistant` (and `tool` for function calling) - Content: The actual message text between the role declaration and `<|im_end|>`

Dual-Mode Reasoning Support

SmolLM3's is a new category of models that can reason, or not. It enables this feature through special formatting and a parameter. If the parameter is set to `think`, the model will show its reasoning process. This is communicated to the model through the `thinking` token.

Standard Mode (`no_think`):

```
<|im_start|>user
What is 15 × 24?<|im_end|>
<|im_start|>assistant
15 × 24 = 360<|im_end|>
```

Thinking Mode (`think`):

```
<|im_start|>user
What is 15 × 24?<|im_end|>
<|im_start|>assistant
<|thinking|>
I need to multiply 15 by 24. Let me break this down:
15 × 24 = 15 × (20 + 4) = (15 × 20) + (15 × 4) = 300 + 60 = 360
</|thinking|>
```

```
15 × 24 = 360<|im_end|>
```

This dual-mode capability allows SmolLM3 to show its reasoning process when needed, making it perfect for combining complex and simple tasks.

Working with SmolLM3 Chat Templates in Code

The `transformers` library automatically handles chat template formatting through the tokenizer. This means you only need to structure your messages correctly, and the library takes care of the special token formatting. Here's how to work with SmolLM3's chat template:

```
from transformers import AutoTokenizer

# Load SmolLM3's tokenizer
tokenizer = AutoTokenizer.from_pretrained("HuggingFaceTB/SmolLM3-3B")

# Structure your conversation as a list of message dictionaries
messages = [
    ,
    ,

]

# Apply the chat template
formatted_chat = tokenizer.apply_chat_template(
    messages,
    tokenize=False,  # Return string instead of tokens
    add_generation_prompt=True  # Add prompt for next assistant response
)

print(formatted_chat)
```

Output:

```
<|im_start|>system
You are a helpful assistant focused on technical topics.<|im_end|>
<|im_start|>user
Can you explain what a chat template is?<|im_end|>
<|im_start|>assistant
A chat template structures conversations between users and AI models by providing a
consistent format that helps the model understand different roles and maintain
context.<|im_end|>
<|im_start|>assistant
```

Understanding the Message Structure

Each message in the conversation follows a simple dictionary format:

- `role`: Identifies who is speaking (`system`, `user`, `assistant`, or `tool`).
- `content`: The actual message content.

Message Types:

1. **System Messages**: Set behavior and context for the entire conversation
2. **User Messages**: Questions, requests, or statements from the human user
3. **Assistant Messages**: Responses from the AI model
4. **Tool Messages**: Results from function calls (for advanced use cases)

System Messages: Setting the Context

System messages are crucial for controlling SmolLM3's behavior. They act as persistent instructions that influence all subsequent interactions. To create a system message, you can use the `system` role and the `content` key:

```
# Professional assistant
system_message =

# Technical expert
system_message =

# Creative assistant
system_message =
```

> [!TIP] System messages have a significant impact on the model's behavior. They are the first message in the conversation and they set the tone for the entire conversation. They should be specific, set boundaries, provide context, and use examples.

Multi-Turn Conversations

SmolLM3 can maintain context across multiple conversation turns. Each message builds upon the previous context. For example, the following code creates a conversation with a helpful programming tutor:

```
conversation = [
    ,
    ,
    ,
    ,
    !'\n\nresult = greet('Alice')\nprint(result)  # Output: Hello, Alice!\n```"},
    ,
]
```

Generation Prompts: Controlling Model Behavior

One of the most important concepts in chat templates is the **generation prompt**. This tells the model when it should start generating a response versus continuing existing text.

Understanding add_generation_prompt

The `add_generation_prompt` parameter controls whether the template adds tokens that indicate the start of a bot response:

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("HuggingFaceTB/SmolLM3-3B")

messages = [
    ,
    ,

]

# Without generation prompt - for completed conversations
formatted_without = tokenizer.apply_chat_template(
    messages,
    tokenize=False,
    add_generation_prompt=False
)

print("Without generation prompt:")
print(formatted_without)
print("\n" + "="*50 + "\n")

# With generation prompt - for inference
formatted_with = tokenizer.apply_chat_template(
    messages,
    tokenize=False,
    add_generation_prompt=True
)
```

```
print("With generation prompt:")
print(formatted_with)
```

Output:

```
Without generation prompt:
<|im_start|>user
Hi there!<|im_end|>
<|im_start|>assistant
Nice to meet you!<|im_end|>
<|im_start|>user
Can I ask a question?<|im_end|>

================================================

With generation prompt:
<|im_start|>user
Hi there!<|im_end|>
<|im_start|>assistant
Nice to meet you!<|im_end|>
<|im_start|>user
Can I ask a question?<|im_end|>
<|im_start|>assistant
```

The generation prompt ensures that when the model generates text, it will write a bot response instead of doing something unexpected like continuing the user's message.

When to Use Generation Prompts

- For inference: Use `add_generation_prompt=True` when you want the model to generate a response.
- For training: Use `add_generation_prompt=False` when preparing training data with complete conversations.
- For evaluation: Use `add_generation_prompt=True` to test model responses.

Continuing Final Messages: Advanced Response Control

The `continue_final_message` parameter allows you to make the model continue the last message in a conversation instead of starting a new one. This is particularly useful for "prefilling" responses or ensuring specific output formats.

Basic Example

```
# Prefill a JSON response
chat = [
    ,
    ,
]

# Continue the final message
formatted_chat = tokenizer.apply_chat_template(
    chat,
    tokenize=False,
    continue_final_message=True
)

print("Continuing final message:")
print(formatted_chat)
print("\n" + "="*50 + "\n")

# Compare with starting a new message
formatted_new = tokenizer.apply_chat_template(
    chat,
    tokenize=False,
    add_generation_prompt=True
)

print("Starting new message:")
print(formatted_new)
```

Output:

```
Continuing final message:
<|im_start|>user
Can you format the answer in JSON?<|im_end|>
<|im_start|>assistant
,
    ,

]

# The model will continue with just the answer, maintaining JSON structure
```

2. Code Completion:

```
# Guide the model to complete code
messages = [
    ,

]
```

```
# Model will complete the recursive call
```

3. Step-by-Step Reasoning:

```
# Guide the model through structured thinking
messages = [
    ,

]

# Model will continue with the first step
```

Important Notes

- You cannot use `add_generation_prompt=True` and `continue_final_message=True` together
- The final message must have the "assistant" role when using `continue_final_message=True`
- This feature removes end-of-sequence tokens from the final message

Working with Reasoning Mode

SmolLM3's dual-mode reasoning can be controlled through special formatting:

Standard vs Thinking Mode

```
# Standard mode - direct answer
standard_messages = [
    ,

]

# Thinking mode - show reasoning process
thinking_messages = [
    ,

]

# Apply templates
standard_formatted = tokenizer.apply_chat_template(standard_messages,
tokenize=False)
thinking_formatted = tokenizer.apply_chat_template(thinking_messages,
tokenize=False)

print("Standard mode:")
print(standard_formatted)
```

```
print("\nThinking mode:")
print(thinking_formatted)
```

Training with Thinking Mode

When preparing datasets with thinking mode, you can control whether to include the reasoning:

```
def create_thinking_example(question, answer, reasoning=None):
    """Create a training example with optional thinking"""
    if reasoning:
        assistant_content = f"<|thinking|>\n\n</|thinking|>\n\n"
    else:
        assistant_content = answer

    return [
        ,

    ]

# Example usage
math_example = create_thinking_example(
    question="What is the derivative of x²?",
    answer="The derivative of x² is 2x",
    reasoning="Using the power rule: d/dx(x^n) = n·x^(n-1)\nFor x²: n=2, so d/
dx(x²) = 2·x^(2-1) = 2x"
)
```

Tool Usage and Function Calling

Modern chat templates support tool usage and function calling. Here's how to work with tools in SmolLM3:

Defining Tools

```
# Define available tools
tools = [
    ,
                    "unit":
                },
                "required": ["location"]
            }
        }
    },

                },
                "required": ["expression"]
            }
```

```
        }
    }
]
```

Chat Templates with Tools

```
# Conversation with tool usage
messages = [
    ,
    ,
    '
                }
            }
        ]
    },
    '
    },

]

# Apply chat template with tools
formatted_with_tools = tokenizer.apply_chat_template(
    messages,
    tools=tools,
    tokenize=False,
    add_generation_prompt=False
)

print("Chat template with tools:")
print(formatted_with_tools)
```

The output of the chat template with tools is:

```
Chat template with tools:
<|im_start|>system
## Metadata

Knowledge Cutoff Date: June 2025
Today Date: 01 September 2025
Reasoning Mode: /think

## Custom Instructions

You are a helpful assistant with access to tools.

### Tools

You may call one or more functions to assist with the user query.
You are provided with function signatures within <tools></tools> XML tags:

<tools>
, 'unit': }, 'required': ['location']}}}
```

```
}, 'required': ['expression']}}}
</tools>

For each function call, return a json object with function name and arguments
within <tool_call></tool_call> XML tags:
<tool_call>

...
<|im_end|>
<|im_start|>assistant
The weather in Paris is currently sunny with a temperature of 22°C and 60%
humidity. It's a beautiful day!<|im_end|>
```

Training with Tool Usage

```
def format_tool_dataset(examples):
    """Format dataset with tool usage for training"""
    formatted_texts = []

    for messages, tools in zip(examples["messages"], examples.get("tools", [None] *
len(examples["messages"]))):
        formatted_text = tokenizer.apply_chat_template(
            messages,
            tools=tools,
            tokenize=False,
            add_generation_prompt=False
        )
        formatted_texts.append(formatted_text)

    return
```

Advanced Template Customization

For advanced use cases, you might need to customize or understand chat templates more deeply:

Inspecting a Model's Chat Template

```
# View the actual template
print("SmolLM3 Chat Template:")
print(tokenizer.chat_template)

# See what special tokens are used
print("\nSpecial tokens:")
print(f"BOS: ")
print(f"EOS: ")
print(f"UNK: ")
print(f"PAD: ")
```

```
# Check for custom tokens
special_tokens = tokenizer.special_tokens_map
for name, token in special_tokens.items():
    print(f": ")
```

Custom Template Creation

```
# Create a custom template (advanced users only)
custom_template = """









<|system|>}<|end|>



<|user|>}<|end|>

<|assistant|>}<|end|>



<|assistant|>

"""

# Apply custom template (be very careful with this!)
# tokenizer.chat_template = custom_template
```

Template Debugging

```
def debug_chat_template(messages, tokenizer):
    """Debug chat template application"""

    # Apply template
    formatted = tokenizer.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=True
    )

    # Tokenize and decode to see actual tokens
    tokens = tokenizer(formatted, return_tensors="pt")

    print("=== TEMPLATE DEBUG ===")
    print(f"Input messages: ")
    print(f"Formatted length:  chars")
    print(f"Token count: ")
```

```
    print("\nFormatted text:")
    print(repr(formatted))  # Shows escape characters
    print("\nTokens:")
    print(tokens['input_ids'][0].tolist()[:20], "...")  # First 20 tokens
    print("\nDecoded tokens:")
    for i, token_id in enumerate(tokens['input_ids'][0][:20]):
        token = tokenizer.decode([token_id])
        print(f":  -> ")

# Example usage
debug_messages = [
    ,

]

debug_chat_template(debug_messages, tokenizer)
```

Key Takeaways

Understanding chat templates is crucial for effective instruction tuning. Here are the essential points to remember:

Core Concepts

1. **Template Consistency**: Always use the same template format for training and inference - mismatches can significantly hurt performance
2. **Generation Prompts**: Use `add_generation_prompt=True` for inference, `False` for training data preparation
3. **Role Structure**: Clear role definitions (`system`, `user`, `assistant`, `tool`) help models understand conversation flow
4. **Context Management**: Leverage SmolLM3's extended context window efficiently by managing conversation history
5. **Special Token Handling**: Let templates handle special tokens - avoid adding them manually

Advanced Features

1. **Dual-Mode Reasoning**: Use `<|thinking|>` tags for complex problems requiring step-by-step reasoning
2. **Message Continuation**: Use `continue_final_message=True` for structured output and prefilling responses
3. **Tool Integration**: Modern templates support function calling and tool usage for enhanced capabilities

4. **Pipeline Automation**: Text generation pipelines handle templates automatically for production use
5. **Multi-Dataset Training**: Standardize different dataset formats before combining for training

Training Best Practices

1. **Dataset Preparation**: Apply templates with `add_generation_prompt=False` and `add_special_tokens=False` for training
2. **Quality Control**: Debug templates thoroughly to ensure proper formatting
3. **Performance Monitoring**: Incorrect template usage can significantly impact model performance
4. **Multimodal Support**: Templates extend to vision and audio models with appropriate modifications

Common Pitfalls to Avoid

- Template mismatch: Using a different template than the model was trained on.
- Double special tokens: Adding special tokens when the template already includes them.
- Missing system messages: Not providing enough context for consistent model behavior.
- Inconsistent formatting: Mixing different conversation formats in the same dataset.
- Wrong generation prompts: Using incorrect `add_generation_prompt` settings for your use case.
- Ignoring tool syntax: Not properly formatting tool calls and responses.
- Context overflow: Not managing long conversations within token limits.

Production Considerations

- Pipeline usage: Use automated pipelines for consistent template application in production.
- Error handling: Implement validation for message formats and role sequences.
- Performance optimization: Cache formatted templates when possible for repeated use.
- Monitoring: Track template application success rates and formatting consistency.

- Version control: Maintain template versions alongside model versions for reproducibility.

Beyond Basic Templates: Advanced Topics

This guide covered the fundamentals, but chat templates support many advanced features:

- Multimodal templates: Handling images, audio, and video in conversations.
- Document integration: Including external documents and knowledge bases.
- Custom template creation: Building specialized templates for domain-specific applications.
- Template optimization: Performance tuning for high-throughput applications.

For these advanced topics, refer to the specialized documentation linked below.

Next Steps

Now that you have a comprehensive understanding of chat templates, you're ready to learn about supervised fine-tuning, where we'll use these templates to train SmolLM3 on custom datasets.

Next: Supervised Fine-Tuning

Comprehensive Resources and Further Reading

Official Documentation

- Getting Started with Chat Templates - Basic concepts and usage patterns
- Multimodal Chat Templates - Vision and audio integration
- Tools and Documents - Function calling and document integration
- Advanced Usage - Custom templates and optimization

Model and Dataset Resources

- SmolLM3 Model Card - Official model documentation and usage examples
- SmolTalk2 Dataset - Training dataset used for SmolLM3 with template examples
- TRL Documentation - Training framework with chat template integration

Technical References

- Transformers Chat Templating API - Complete API reference
- Jinja2 Template Engine - Template syntax and advanced features
- OpenAI Chat Completions API - Industry standard message format

Community Resources

- Hugging Face Forum - Community discussions and troubleshooting
- Discord Server - Real-time help and community interaction
- GitHub Issues - Bug reports and feature requests

Supervised Fine-Tuning with SmolLM3

Supervised Fine-Tuning (SFT) is the cornerstone of instruction tuning - it's how we transform a base language model into an instruction-following assistant. In this section, you'll learn to fine-tune **SmolLM3** using real-world datasets and production-ready tools.

What is Supervised Fine-Tuning?

SFT is the process of continuing to train a pre-trained model on task-specific datasets with labeled examples. Think of it as specialized education:

- Pre-training teaches the model general language understanding (like learning to read).
- Supervised fine-tuning teaches specific skills and behaviors (like learning to do a specific task).

The key insight behind SFT is that we're not teaching the model new knowledge from scratch. Instead, we're **reshaping how existing knowledge is applied**. The pre-trained model already understands language, grammar, and has absorbed vast amounts of factual information. SFT focuses this general capability toward specific application patterns, response styles, and task-specific requirements.

This approach is effective because it leverages the rich representations learned during pre-training while requiring significantly less computational resources than training from scratch. The model learns to recognize instruction patterns, maintain

conversation context, follow safety guidelines, and generate responses in desired formats.

> [!TIP] Before starting SFT, consider whether using an existing instruction-tuned model with well-crafted prompts would suffice for your use case. SFT involves significant computational resources and engineering effort, so it should only be pursued when prompting existing models proves insufficient. Learn more about this decision process in the Hugging Face LLM Course.

The SmolLM3 SFT Journey

SmolLM3's instruction-following capabilities come from a sophisticated SFT process:

1. **Base Model** (`SmolLM3-3B-Base`): Trained on 11T tokens of general text
2. **SFT Training**: Fine-tuned on curated instruction datasets including SmolTalk2
3. **Preference Alignment**: Further refined using techniques like APO (Anchored Preference Optimization)

This multi-stage approach creates a model that's both knowledgeable and helpful.

Why SFT Works: The Science Behind It

SFT is effective because it leverages the rich representations learned during pre-training while adapting the model's behavior patterns. During SFT, the model's parameters are fine-tuned through gradient descent on task-specific examples, causing subtle but important changes in how the model processes and generates text.

Specifically, the process works through several key mechanisms:

Behavioral Adaptation: The model learns to recognize instruction patterns and respond appropriately. This involves updating the attention mechanisms to focus on instruction cues in language and adjusting the output distribution to favor the desired responses. Research has shown that instruction tuning primarily affects the model's surface-level behavior rather than its underlying knowledge (Wei et al., 2021).

Task Specialization: Rather than learning entirely new concepts, the model learns to apply its existing knowledge in specific contexts. This is why SFT is much more efficient than pre-training - we're refining existing capabilities rather than building

them from scratch. Studies indicate that most of the factual knowledge comes from pre-training, while SFT teaches the model how to format and present this knowledge appropriately (Ouyang et al., 2022).

Safety Alignment: Through exposure to carefully curated examples, the model learns to be more helpful, harmless, and honest. This involves both learning what to say and what not to say in various situations. The effectiveness of this approach has been demonstrated in works like InstructGPT (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022).

> [!TIP] SFT doesn't teach new facts - it teaches new behaviors. The model already knows about the world from pre-training; SFT teaches it how to be a helpful assistant using that knowledge.

The mathematical foundation involves minimizing the cross-entropy loss between the model's predictions and the target responses in your training dataset. This process gradually shifts the model's probability distributions to favor the types of responses demonstrated in your training examples.

When to Use Supervised Fine-Tuning

The key question is: "Does my use case require behavior that differs significantly from general-purpose conversation?" If yes, SFT is likely beneficial.

Decision framework: Use this checklist to determine if SFT is appropriate for your project: - Have you tried prompt engineering with existing instruction-tuned models? - Do you need consistent output formats that prompting cannot achieve? - Is your domain specialized enough that general models struggle? - Do you have high-quality training data (at least 1,000 examples)? - Do you have the computational resources for training and evaluation?

If you answered "yes" to most of these, SFT is likely worth pursuing.

The SFT Process

Now let's move on to the process of SFT itself. The SFT process follows a systematic approach that ensures high-quality results:

1. Dataset Preparation and Selection

The quality of your training data is the most critical factor for successful SFT. Unlike pre-training where quantity often matters most, SFT prioritizes quality and relevance. Your dataset should contain input-output pairs that demonstrate exactly the behavior you want your model to learn.

Choose the Right Dataset: - SmolTalk2: The dataset used to train SmolLM3, containing high-quality instruction-response pairs. - Domain-specific datasets: For specialized applications (medical, legal, technical). - Custom datasets: Your own curated examples for specific use cases.

Each training example should consist of: 1. **Input prompt**: The user's instruction or question 2. **Expected response**: The ideal assistant response 3. **Context** (optional): Any additional information needed

> [!TIP] Dataset size guidelines: - Minimum: 1,000 high-quality examples for basic fine-tuning. - Recommended: 10,000+ examples for robust performance. - Quality over quantity: 1,000 well-curated examples often outperform 10,000 mediocre ones.
>
> Remember: Your model will learn to mimic the patterns in your training data, so invest time in data curation.

2. Environment Setup and Configuration

To set up an environment for SFT, we will need advance compute resources. We have three main options:

1. **Local GPU**: If you are lucky enough to have a access to a GPU with (at least 16GB of VRAM), you can train your model locally!
2. **Hugging Face Jobs**: If you don't have a GPU and don't want to use a cloud provider, you can use Hugging Face Jobs! We'll go into more detail about this in the next section.
3. **Notebook GPUs**: If you like to use a notebook provider like Google Colab, you can use their GPUs!
4. **Cloud GPU**: If you want to take control of your compute resources, you can use a cloud provider like AWS, GCP, or Azure.

In terms of hardware requirements, you will need a GPU with at least 16GB of VRAM, for example an Nvidia RTX 4080 or A10G.

3. Training Configuration

Choosing the right hyperparameters is crucial for successful SFT. The goal is to find the sweet spot where the model learns effectively without overfitting or becoming unstable. Here's a detailed breakdown of each parameter and how to choose them:

Key Hyperparameters:

Learning Rate (5e-5 to 1e-4): Controls how much the model weights change with each update - Start with 5e-5 for SmolLM3; this is conservative and stable. - Too high: The model becomes unstable; loss oscillates or explodes. - Too low: The model learns very slowly and may not converge in reasonable time.

Batch Size (4-16): Number of examples processed simultaneously - Larger batches: More stable gradients, but require more GPU memory. - Smaller batches: Less memory usage, but noisier gradients. - Use gradient accumulation to achieve larger effective batch sizes.

Max Sequence Length (2048-4096): Maximum tokens per training example - Longer sequences: Can handle more complex conversations. - Shorter sequences: Faster training, less memory usage. - Match your use case: Use the typical length of your target conversations.

Training Steps (1000-5000): Total number of parameter updates - Depends on dataset size: More data usually requires more steps. - Monitor validation loss: Stop when it stops improving. - Rule of thumb: Three to five epochs through your dataset.

Warmup Steps (10% of total): Gradual learning rate increase at start - Prevents early instability: Helps the model adapt gradually. - Typical range: 100-500 steps for most SFT tasks.

> [!TIP] Hyperparameter starting points for SmolLM3:
>
> To bootstrap your training, you can use the following hyperparameters:
>
> **Learning Rate**:
>
> ```python
>
> # Conservative (stable, slower)
>
> learning_rate = 5e-5
> ```

Balanced (recommended)

learning_rate = 1e-4

Aggressive (faster, less stable)

learning_rate = 2e-4 ```

Batch Size:

We can reduce GPU device batch size by using gradient accumulation.

```python
```

Limited GPU Memory

per_device_train_batch_size = 2 gradient_accumulation_steps = 8

Balanced GPU Memory

per_device_train_batch_size = 4 gradient_accumulation_steps = 4

More GPU Memory

per_device_train_batch_size = 8 gradient_accumulation_steps = 2 ```

Max Sequence Length:

```python
```

Very short sequences

max_length = 512

Short sequences

max_length = 1024

Long sequences

max_length = 2048

> # Very long sequences
>
> max_length = 4096 ```

4. Monitoring and Evaluation

Effective monitoring is crucial for successful SFT. Unlike pre-training where you primarily watch loss decrease, SFT requires careful attention to both quantitative metrics and qualitative outputs. The goal is to ensure your model is learning the desired behaviors without overfitting or developing unwanted patterns.

Key Metrics to Monitor:

Training Loss: Should decrease steadily but not too rapidly - Healthy pattern: Smooth, gradual decrease. - Warning signs: Sudden spikes, oscillations, or plateaus. - Typical range: Starts around 2-4, should decrease to 0.5-1.5.

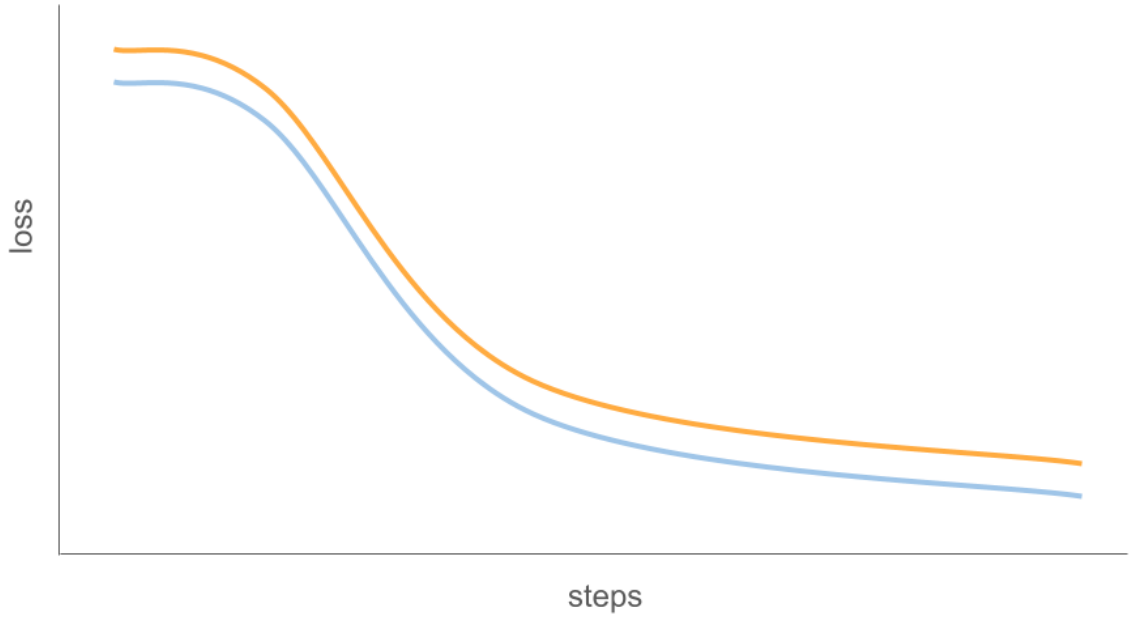
Validation Loss: Most important metric for preventing overfitting - Should track training loss: A small gap indicates good generalization. - Growing gap: Sign of overfitting; the model may be memorizing training data. - Use for early stopping: Stop training when validation loss stops improving.

Sample Outputs: Regular qualitative checks are essential - Generate responses: Test the model on held-out prompts during training. - Check format consistency: Ensure the model follows desired response patterns. - Monitor for degradation: Watch for repetitive or nonsensical outputs.

Resource Usage: Track GPU memory and training speed - Memory spikes: May indicate batch size is too large. - Slow training: Could suggest inefficient data loading or processing.

Understanding Loss Patterns in SFT

Training loss typically follows three distinct phases, as illustrated in this example from the Hugging Face LLM Course:



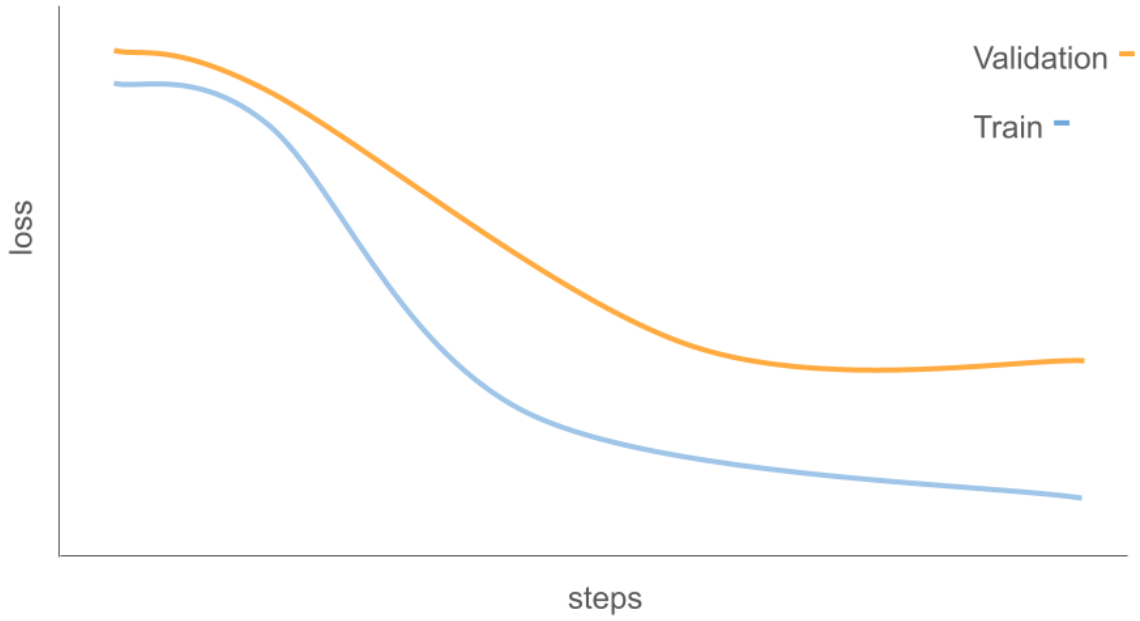
1. **Initial Sharp Drop**: Rapid adaptation to new data distribution
2. **Gradual Stabilization**: Learning rate slows as model fine-tunes
3. **Convergence**: Loss values stabilize, indicating training completion

Healthy Training Pattern: The key indicator of successful training is a small gap between training and validation loss, suggesting the model is learning generalizable patterns rather than memorizing specific examples.

Warning Signs to Watch For

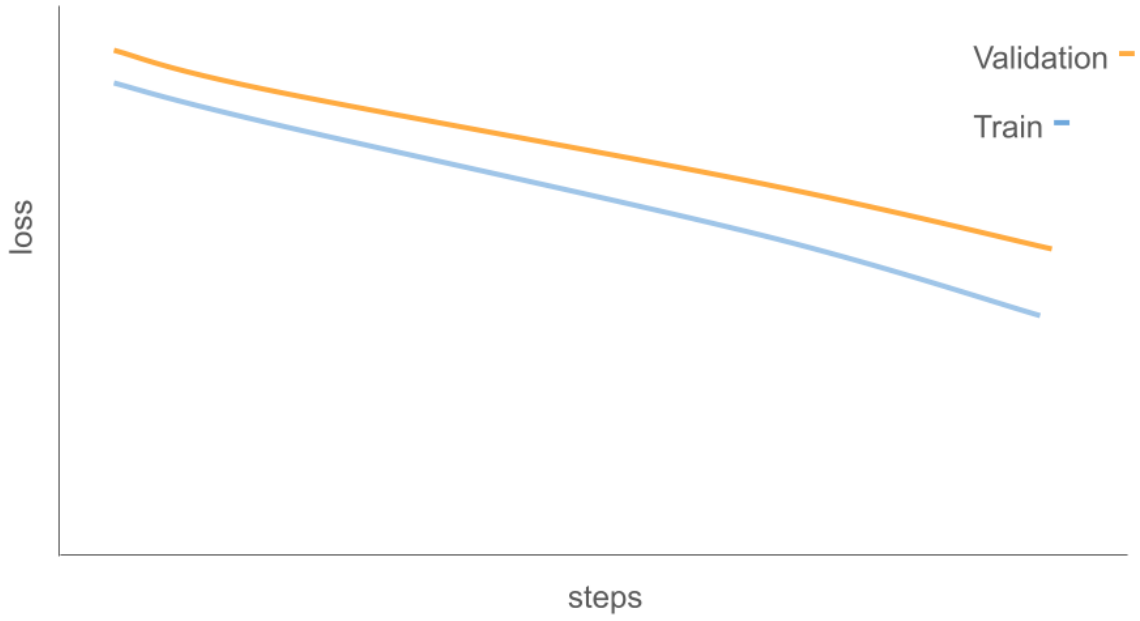
Several patterns in the loss curves can indicate potential issues:

Overfitting Pattern



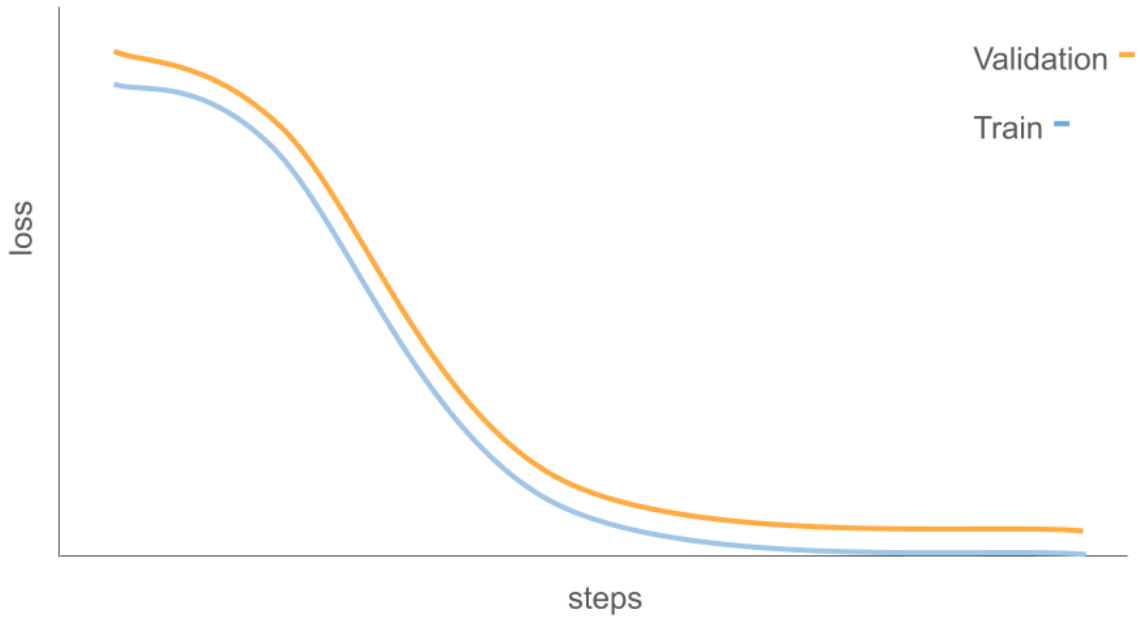
If validation loss increases while training loss continues to decrease, your model is overfitting. Consider: - Reducing training steps or epochs - Increasing dataset size or diversity - Adding regularization techniques - Using early stopping based on validation loss

Underfitting Pattern



If loss doesn't show significant improvement, the model might be: - Learning too slowly (try increasing learning rate) - Struggling with task complexity (check data quality) - Hitting architectural limitations (consider different model size)

Potential Memorization



Extremely low loss values could suggest memorization rather than learning. This is concerning if: - Model performs poorly on new, similar examples - Outputs lack diversity or creativity - Responses are too similar to training examples

> [!TIP] Learn more about loss interpretation in the Hugging Face LLM Course.

Experiment Tracking with Trackio: For comprehensive experiment tracking, we recommend **Trackio** - a lightweight, free experiment tracking library built on Hugging Face infrastructure. Trackio provides:

- Drop-in replacement: API compatible with `wandb.init`, `wandb.log`, and `wandb.finish`.
- Local-first design: Dashboard runs locally by default, with optional Hugging Face Spaces hosting.
- Free hosting: Everything, including hosting on Hugging Face Spaces, is free.
- Lightweight: Fewer than 3,000 lines of Python code, easily extensible.

We can track any metrics during training, for example:

```
# Simple Trackio integration


# Initialize tracking
trackio.init(project="smollm3-sft")

# Log metrics during training
trackio.log()

# Finish tracking
trackio.finish()
```

The most convenient way to track your training is to use trackio's `transformers` integration. You can specify your Trackio project name and space ID using environment variables:

```
export TRACKIO_PROJECT_NAME="my-project"
export TRACKIO_SPACE_ID="username/space_id"
```

Or you can set them in your code:

```
os.environ["TRACKIO_PROJECT_NAME"] = "my-project"
os.environ["TRACKIO_SPACE_ID"] = "username/space_id"
```

Then you can use the `SFTTrainer` class from TRL to track your training and let it handle the tracking for you.

```
from trl import SFTTrainer

trainer = SFTTrainer(
    model=model,
    train_dataset=dataset["train"],
    args=config,
)
```

Trackio will serve an application with the metrics from training that looks like this:

Logged metrics

While training and evaluating we record the following reward metrics:

- `global_step`: The total number of optimizer steps taken so far.
- `epoch`: The current epoch number, based on dataset iteration.

- `num_tokens`: The total number of tokens processed so far.
- `loss`: The average cross-entropy loss computed over non-masked tokens in the current logging interval.
- `entropy`: The average entropy of the model's predicted token distribution over non-masked tokens.
- `mean_token_accuracy`: The proportion of non-masked tokens for which the model's top-1 prediction matches the ground truth token.
- `learning_rate`: The current learning rate, which may change dynamically if a scheduler is used.
- `grad_norm`: The L2 norm of the gradients, computed before gradient clipping.

Expected dataset type and format

SFT supports both language modeling and prompt-completion datasets. The [`SFTTrainer`] is compatible with both standard and conversational dataset formats. When provided with a conversational dataset, the trainer will automatically apply the chat template to the dataset.

```
# Standard language modeling


# Conversational language modeling
,
              ]}

# Standard prompt-completion


# Conversational prompt-completion
],
 "completion": []}
```

If your dataset is not in one of these formats, you can preprocess it to convert it into the expected format. Here is an example with the FreedomIntelligence/medical-o1-reasoning-SFT dataset:

```
from datasets import load_dataset

dataset = load_dataset("FreedomIntelligence/medical-o1-reasoning-SFT", "en")

def preprocess_function(example):
    return ],
        "completion": [
            </think>"}
```

```
        ],
    }

dataset = dataset.map(preprocess_function, remove_columns=["Question", "Response",
"Complex_CoT"])
print(next(iter(dataset["train"])))
```

```
    ],
    "completion": [

    ],
}
```

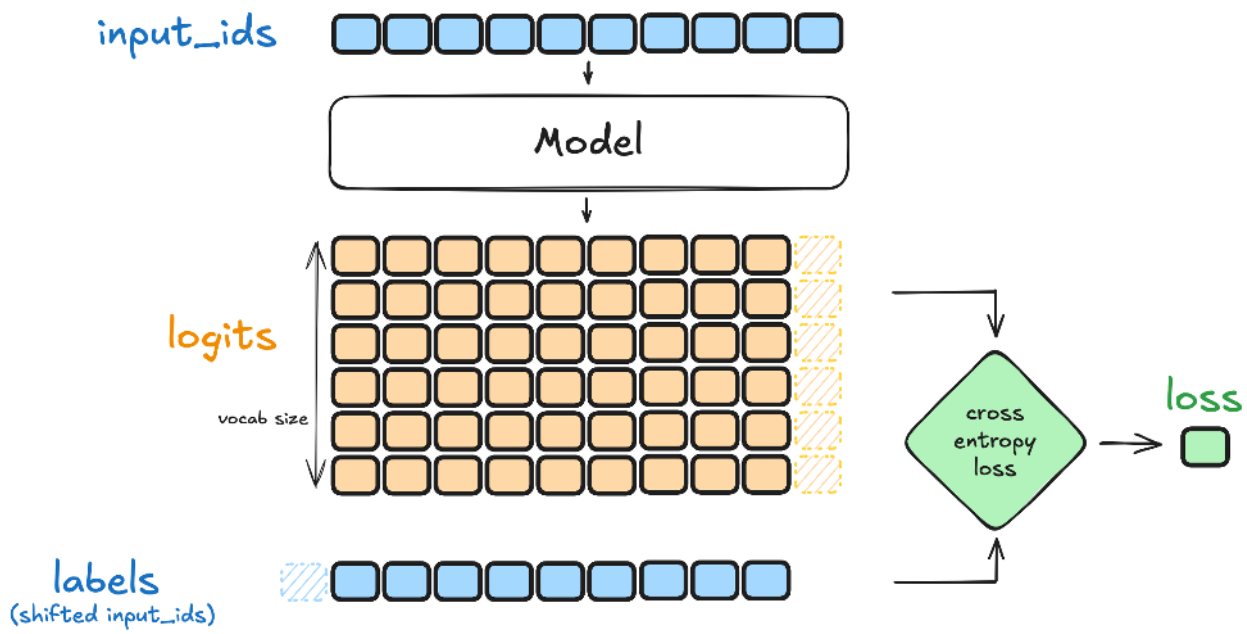
Chat Templates in Training

We'll return briefly to chat templates in the context of training. Using chat templates correctly during training is crucial for model performance. Here are the key considerations and best practices:

Preprocessing and tokenization

During training, each example is expected to contain a **text field** or a **(prompt, completion)** pair, depending on the dataset format. For more details on the expected formats, see Dataset formats. The `SFTTrainer` tokenizes each input using the model's tokenizer. If both prompt and completion are provided separately, they are concatenated before tokenization.

Computing the loss



The loss used in SFT is the **token-level cross-entropy loss**, defined as:

$$ \mathcal_}(\theta) = - \sum_^ \log p_\theta(y_t \mid y_), $$

where \(y_t \) is the target token at timestep \(t \), and the model is trained to predict the next token given the previous ones. In practice, padding tokens are masked out during loss computation.

Supervised Fine-Tuning with TRL (Transformer Reinforcement Learning)

TRL is the go-to toolkit for training language models, built specifically for instruction tuning and alignment. It's what we'll use throughout this course.

Why TRL?

- Production ready: Used by major organizations and research labs.
- Comprehensive: Supports SFT, DPO, ORPO, PPO, and more advanced techniques.
- Efficient: Optimized for memory usage and training speed.
- Flexible: Works with any Hugging Face model.
- CLI support: Command-line tools for scalable training workflows.

Key Components

- **SFTTrainer**: The core class for supervised fine-tuning
- **SFTConfig**: Configuration management for training parameters
- **CLI Tools**: Command-line interface for production workflows
- **Integration**: Seamless integration with Hugging Face Hub, Trackio, Weights & Biases, and more

TRL's Architecture

TRL is built on top of the Hugging Face ecosystem: - Transformers: Model loading and inference. - Datasets: Data processing and management. - Accelerate: Distributed training and optimization. - PEFT: Parameter-efficient fine-tuning (LoRA, QLoRA).

This integrated approach means you get all the benefits of the Hugging Face ecosystem while using state-of-the-art training techniques.

> [!TIP] TRL versus other training libraries: - TRL: Specialized for LLM training, built for instruction tuning. - Transformers Trainer: General purpose, suitable for basic fine-tuning. - DeepSpeed: Focuses on large-scale distributed training. - Accelerate: Provides low-level distributed training primitives.
>
> TRL provides the best balance of ease-of-use and advanced features for SFT. For more details on training approaches, see the Hugging Face LLM Course.

Hands-On: Your First SmolLM3 Fine-Tune

Ready to put theory into practice? Here's a preview of what you'll build in the exercises. You can use either Python or CLI approach:

```
from transformers import AutoModelForCausalLM, AutoTokenizer
from trl import SFTTrainer, SFTConfig
from datasets import load_dataset


# Initialize experiment tracking
wandb.init(project="smollm3-sft", name="my-first-sft-run")

# Load SmolLM3 base model
model = AutoModelForCausalLM.from_pretrained("HuggingFaceTB/SmolLM3-3B-Base")
tokenizer = AutoTokenizer.from_pretrained("HuggingFaceTB/SmolLM3-3B-Base")
```

```
# Load SmolTalk2 dataset
dataset = load_dataset("HuggingFaceTB/smoltalk2_everyday_convs_think")

# Configure training with Trackio integration
config = SFTConfig(
    output_dir="./smollm3-finetuned",
    per_device_train_batch_size=4,
    learning_rate=5e-5,
    max_steps=1000,
    report_to="trackio",  # Enable Trackio logging
)

# Train!
trainer = SFTTrainer(
    model=model,
    train_dataset=dataset["train"],
    args=config,
)
trainer.train()
```

```
# Fine-tune SmolLM3 using TRL CLI with Trackio tracking
trl sft \
    --model_name_or_path HuggingFaceTB/SmolLM3-3B-Base \
    --dataset_name HuggingFaceTB/smoltalk2_everyday_convs_think \
    --output_dir ./smollm3-sft-model \
    --per_device_train_batch_size 4 \
    --learning_rate 5e-5 \
    --max_steps 1000 \
    --logging_steps 50 \
    --save_steps 200 \
    --report_to trackio \
    --push_to_hub \
    --hub_model_id your-username/smollm3-custom
```

Severless Training Options

While you can train models locally, cloud infrastructure offers significant advantages for SFT training. For users who want to skip the complexity of GPU setup and environment management, **Hugging Face Jobs** provides a seamless solution.

See Training with Hugging Face Jobs for fully managed cloud infrastructure with high-end GPUs, automatic scaling, and integrated monitoring.

Key Takeaways

1. **SFT is Essential**: It's the bridge between base models and instruction-following assistants
2. **Data Quality Matters**: High-quality datasets lead to better fine-tuned models - invest time in curation
3. **Monitor Carefully**: Watch both loss curves and actual outputs to catch issues early
4. **TRL Simplifies Everything**: From research to production, TRL provides the tools you need
5. **SmolLM3 is Perfect for Learning**: Powerful enough to be useful, small enough to be accessible
6. **Multiple Approaches**: Both programmatic and CLI workflows for different use cases

> [!TIP] 🎓 **Continue Learning**: This introduction covers the fundamentals, but SFT is a deep topic. For more advanced techniques, evaluation methods, and troubleshooting tips, explore the Hugging Face LLM Course which provides comprehensive coverage of modern LLM training techniques.

Next Steps

Now that you understand the theory, choose your training approach:

Training with Hugging Face Jobs - Use cloud infrastructure for training Hands-On Exercises - Fine-tune your own SmolLM3 model locally or in the cloud

Resources and Further Reading

- Training with Hugging Face Jobs - Cloud-based training with managed infrastructure
- Trackio Documentation - Free, lightweight experiment tracking
- TRL Documentation - Comprehensive guide to all TRL features
- SFTTrainer API Reference - Detailed parameter documentation
- SmolTalk2 Dataset - The dataset that trained SmolLM3
- SmolLM3 Model Card - Official model documentation

- TRL CLI Documentation - Command-line interface guide

LoRA and PEFT: Efficient Fine-Tuning

Parameter-Efficient Fine-Tuning (PEFT) lets you adapt large models by training a small number of additional parameters while keeping the base model frozen. The most widely used PEFT method is LoRA (Low-Rank Adaptation), which injects trainable low-rank updates into linear layers. This often reduces trainable parameters by ~90% while preserving performance.

When to use PEFT

- You have limited compute or memory budget
- You need to quickly adapt a base model to multiple tasks/domains
- You want fast iteration and small artifacts (adapter weights are usually a few MB)

Understanding LoRA

LoRA has become the most widely adopted PEFT method. It works by adding small rank decomposition matrices to the attention weights, typically reducing trainable parameters by about 90%.

LoRA (Low-Rank Adaptation) is a parameter-efficient fine-tuning technique that freezes the pre-trained model weights and injects trainable rank decomposition matrices into the model's layers. Instead of training all model parameters during fine-tuning, LoRA decomposes the weight updates into smaller matrices through low-rank decomposition, significantly reducing the number of trainable parameters while maintaining model performance. For example, when applied to GPT-3 175B, LoRA reduced trainable parameters by 10,000x and GPU memory requirements by 3x compared to full fine-tuning. You can read more about LoRA in the LoRA paper.

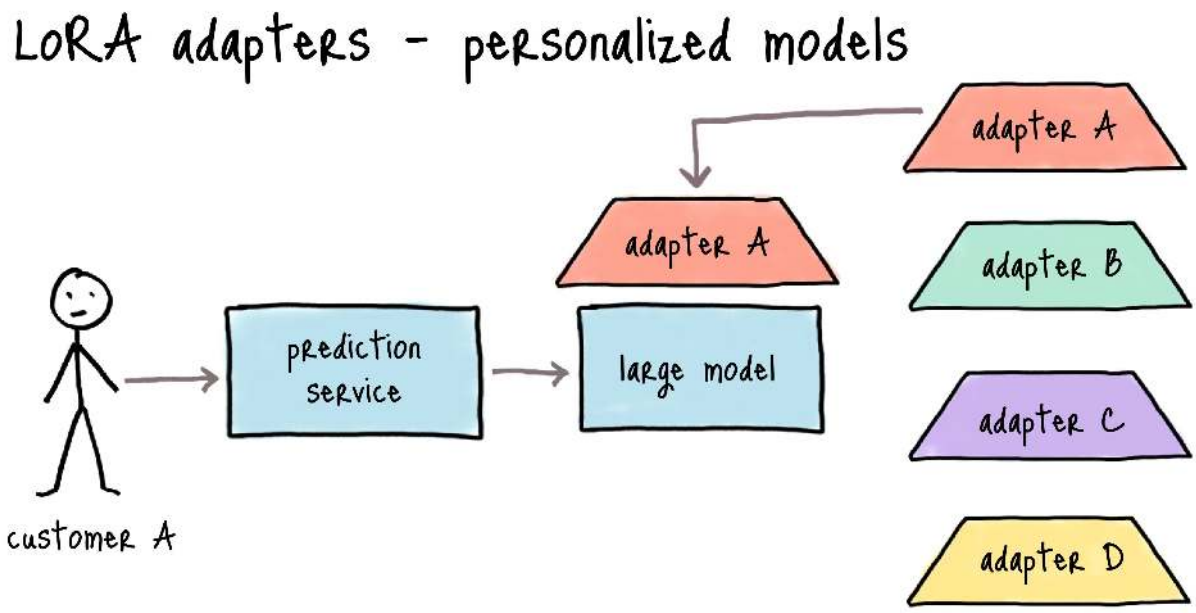
LoRA works by adding pairs of rank decomposition matrices to transformer layers, typically focusing on attention weights. During inference, these adapter weights can be merged with the base model, resulting in no additional latency overhead. LoRA is particularly useful for adapting large language models to specific tasks or domains while keeping resource requirements manageable.

Loading LoRA Adapters

Adapters can be loaded onto a pretrained model with load_adapter(), which is useful for trying out different adapters whose weights aren't merged. Set the active adapter weights with the set_adapter() function. To return the base model, you could use unload() to unload all of the LoRA modules. This makes it easy to switch between different task-specific weights.

```
from transformers import AutoModelForCausalLM
from peft import PeftModel

base_model = AutoModelForCausalLM.from_pretrained("<base_model_name>")
peft_model_id = "<peft_adapter_id>"
model = PeftModel.from_pretrained(base_model, peft_model_id)
```



Merging LoRA Adapters

After training with LoRA, you might want to merge the adapter weights back into the base model for easier deployment. This creates a single model with the combined weights, eliminating the need to load adapters separately during inference.

The merging process requires attention to memory management and precision. Since you'll need to load both the base model and adapter weights simultaneously, ensure sufficient GPU/CPU memory is available. Using `device_map="auto"` in `transformers` will help with automatic memory management. Maintain consistent

precision (e.g., float16) throughout the process, matching the precision used during training and saving the merged model in the same format for deployment. Before deploying, always validate the merged model by comparing its outputs and performance metrics with the adapter-based version.

Adapters are also be convenient for switching between different tasks or domains. You can load the base model and adapter weights separately. This allows for quick switching between different task-specific weights.

> [!TIP] When implementing PEFT methods, start with small rank values (4-8) for LoRA and monitor training loss. Use validation sets to prevent overfitting and compare results with full fine-tuning baselines when possible. The effectiveness of different methods can vary by task, so experimentation is key.

OLoRA

OLoRA utilizes QR decomposition to initialize the LoRA adapters. OLoRA translates the base weights of the model by a factor of their QR decompositions, i.e., it mutates the weights before performing any training on them. This approach significantly improves stability, accelerates convergence speed, and ultimately achieves superior performance.

Using TRL with PEFT

PEFT methods can be combined with TRL (Transformers Reinforcement Learning) for efficient fine-tuning. This integration is particularly useful for RLHF (Reinforcement Learning from Human Feedback) as it reduces memory requirements.

```
from peft import LoraConfig
from trl import SFTTrainer

# Load model with PEFT config
lora_config = LoraConfig(
    r=16,
    lora_alpha=32,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM"
)

trainer = SFTTrainer(
    model="your-model-name",
```

```
    train_dataset=dataset["train"]
    peft_config=lora_config
)
```

Basic Merging Implementation

After training a LoRA adapter, you can merge the adapter weights back into the base model. Here's how to do it:

```
from transformers import AutoModelForCausalLM
from peft import PeftModel

# 1. Load the base model
base_model = AutoModelForCausalLM.from_pretrained(
    "base_model_name",
    dtype=torch.bfloat16,
    device_map="auto"
)

# 2. Load the PEFT model with adapter
peft_model = PeftModel.from_pretrained(
    base_model,
    "path/to/adapter",
    dtype=torch.bfloat16
)

# 3. Merge adapter weights with base model
try:
    merged_model = peft_model.merge_and_unload()
except RuntimeError as e:
    print(f"Merging failed: ")
    # Implement fallback strategy or memory optimization

# 4. Save the merged model
merged_model.save_pretrained("path/to/save/merged_model")
```

If you encounter size discrepancies in the saved model, ensure you're also saving the tokenizer:

```
# Save both model and tokenizer
tokenizer = AutoTokenizer.from_pretrained("base_model_name")
merged_model.save_pretrained("path/to/save/merged_model")
tokenizer.save_pretrained("path/to/save/merged_model")
```

Quick start with TRL + LoRA

The TRL `SFTTrainer` integrates natively with PEFT. Define a `LoraConfig`, pass it to the trainer, and train only the adapter weights.

```
from peft import LoraConfig
from trl import SFTTrainer, SFTConfig

# 1) Configure LoRA
peft_config = LoraConfig(
    r=8,
    lora_alpha=16,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
)

# 2) Create trainer (example)
trainer = SFTTrainer(
    model=model,
    args=SFTConfig(output_dir="lora-adapter", num_train_epochs=1,
per_device_train_batch_size=2, packing=True),
    train_dataset=dataset["train"],
    peft_config=peft_config,
)
trainer.train()
```

After training, you can either:

- Load adapters at inference time alongside the base model, or
- Merge adapters into the base model for simplified deployment.

Resources

- LoRA: Low-Rank Adaptation of Large Language Models
- PEFT Documentation
- Hugging Face blog post on PEFT

Hands-On Exercises: Fine-Tuning SmolLM3

Welcome to the practical section! Here you'll apply everything you've learned about chat templates and supervised fine-tuning using SmolLM3. These exercises progress from basic concepts to advanced techniques, giving you real-world experience with instruction tuning.

Learning Objectives

By completing these exercises, you will: - Master SmolLM3's chat template system - Fine-tune SmolLM3 on real datasets using both Python APIs and CLI tools - Work with the SmolTalk2 dataset that was used to train the original model - Compare base model vs fine-tuned model performance - Deploy your models to Hugging Face Hub - Understand production workflows for scaling fine-tuning

---

Exercise 1: Exploring SmolLM3's Chat Templates

Objective: Understand how SmolLM3 handles different conversation formats and reasoning modes.

SmolLM3 is a hybrid reasoning model which can follow instructions or generated tokens that 'reason' on a complex problem. When post-trained effectively, the model will reason on hard problems and generate direct responses on easy problems.

Environment Setup

> [!WARNING] - You need a GPU with at least 8GB VRAM for training. CPU/MPS can run formatting and dataset exploration, but training larger models will likely fail. - First run will download several GB of model weights; ensure 15GB+ free disk and a stable connection. - If you need access to private repos, authenticate with Hugging Face Hub via `login()`.

Let's start by setting up our environment.

```
# Install required packages (run in Colab or your environment)
pip install "transformers>=4.36.0" "trl>=0.7.0" "datasets>=2.14.0" "torch>=2.0.0"
pip install "accelerate>=0.24.0" "peft>=0.7.0" "trackio"
```

Then, let's import the necessary libraries and set up the accelerator device. below we validate whether we're using a Nvidia GPU, an Apple Metal accelerator or the CPU. In reality, we can't train models on the CPU, so we'll use an accelerator.

```
# Import necessary libraries

from transformers import AutoModelForCausalLM, AutoTokenizer, pipeline
from datasets import load_dataset

from typing import Optional, Dict, Any

if torch.cuda.is_available():
    device = "cuda"
    print(f"Using CUDA GPU: ")
    print(f"GPU memory: GB")
elif hasattr(torch.backends, 'mps') and torch.backends.mps.is_available():
    device = "mps"
    print("Using Apple MPS")
else:
    device = "cpu"
    print("Using CPU - you will need to use a GPU to train models")

# Authenticate with Hugging Face (optional, for private models)
from huggingface_hub import login
# login()  # Uncomment if you need to access private models
```

Take a note of the device you're using and your available GPU memory. If this is below 8GB, you will not be able to do some exercises.

Output
```python output Using CUDA GPU: NVIDIA A100-SXM4-40GB GPU memory: 42.5GB
```

```
</details>

### Load SmolLM3 Models

Now let's load the base and instruct models for comparison.

```python
# Load both base and instruct models for comparison
base_model_name = "HuggingFaceTB/SmolLM3-3B-Base"
instruct_model_name = "HuggingFaceTB/SmolLM3-3B"

# Load tokenizers
base_tokenizer = AutoTokenizer.from_pretrained(base_model_name)
instruct_tokenizer = AutoTokenizer.from_pretrained(instruct_model_name)

# Load models (use smaller precision for memory efficiency)
base_model = AutoModelForCausalLM.from_pretrained(
    base_model_name,
    dtype=torch.bfloat16,
    device_map="auto"
)

instruct_model = AutoModelForCausalLM.from_pretrained(
    instruct_model_name,
    dtype=torch.bfloat16,
    device_map="auto"
)

print("Models loaded successfully!")
```

This will download the models and tokenizers to your local machine from the Hugging Face Hub. This includes the model's parameter weights, tokenizer, and other model configuration defined by the model authors.

Output

You should see green bars loading the model weights. This may take a few minutes.

```python output tokenizer_config.json: 50.4k/? [00:00<00:00, 5.09MB/s] tokenizer.json: 100% 17.2M/17.2M [00:02<00:00, 10.7MB/s] special_tokens_map.json: 100% 151/151 [00:00<00:00, 21.5kB/s] tokenizer_config.json: 50.4k/? [00:00<00:00, 5.45MB/s] tokenizer.json: 100% 17.2M/17.2M [00:00<00:00, 472kB/s] special_tokens_map.json: 100% 289/289 [00:00<00:00, 35.0kB/s] chat_template.jinja: 5.60k/? [00:00<00:00, 577kB/s] config.json: 100% 943/943 [00:00<00:00, 121kB/s] model.safetensors.index.json: 26.9k/? [00:00<00:00, 2.81MB/s] Fetching 2 files: 100% 2/2 [00:32<00:00, 32.11s/it] model-00001-of-00002.safetensors: 100% 4.97G/4.97G [00:31<00:00, 247MB/s] model-00002-of-00002.safetensors: 100% 1.18G/1.18G [00:17<00:00, 57.2MB/s] Loading checkpoint shards: 100%
```
```

2/2 [00:01<00:00, 1.18it/s] generation_config.json: 100% 126/126 [00:00<00:00, 17.1kB/s] config.json: 1.92k/? [00:00<00:00, 229kB/s] model.safetensors.index.json: 26.9k/? [00:00<00:00, 3.14MB/s] Fetching 2 files: 100% 2/2 [00:32<00:00, 32.38s/it] model-00002-of-00002.safetensors: 100% 1.18G/1.18G [00:17<00:00, 92.1MB/s] model-00001-of-00002.safetensors: 100% 4.97G/4.97G [00:31<00:00, 182MB/s] Loading checkpoint shards: 100% 2/2 [00:01<00:00, 1.14it/s] generation_config.json: 100% 182/182 [00:00<00:00, 21.0kB/s] Models loaded successfully!

```
</details>

### Explore Chat Template Formatting

Now let's explore the chat template formatting. We will create different types of conversations to test.

```python
# Create different types of conversations to test
conversations = ,
    ],

    "with_system": [
        ,
        ,
    ],

    "multi_turn": [
        ,
        ,
        ,
        ,
    ],

    "reasoning_task": [
        ,
    ]
}

for conv_type, messages in conversations.items():
    print(f"---  ---")

    # Format without generation prompt (for completed conversations)
    formatted_complete = instruct_tokenizer.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=False
    )

    # Format with generation prompt (for inference)
    formatted_prompt = instruct_tokenizer.apply_chat_template(
```
```

```
        messages,
        tokenize=False,
        add_generation_prompt=True
    )

    print("Complete conversation format:")
    print(formatted_complete)
    print("\nWith generation prompt:")
    print(formatted_prompt)
    print("\n" + "="*50 + "\n")
```

Output

```python output --- SIMPLE_QA --- Complete conversation format: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. <|im_start|>user What is machine learning?<|im_end|> With generation prompt: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. <|im_start|>user What is machine learning?<|im_end|> <|im_start|>assistant
==================================================
--- WITH_SYSTEM --- Complete conversation format: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant specialized in explaining technical concepts clearly. <|im_start|>user What is machine learning?<|im_end|> With generation prompt: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant specialized in explaining technical concepts clearly. <|im_start|>user What is machine learning?<|im_end|> <|im_start|>assistant
==================================================
--- MULTI_TURN --- Complete conversation format: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a math tutor. <|im_start|>user What is calculus?<|im_end|> <|im_start|>assistant Calculus is a branch of mathematics that deals with rates of change and accumulation of quantities.<|im_end|> <|im_start|>user Can you give me a simple example?<|im_end|> With generation prompt: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a math tutor. <|im_start|>user What is calculus?<|im_end|> <|im_start|>assistant Calculus is a branch of mathematics that deals with rates of
```

change and accumulation of quantities.<|im_end|> <|im_start|>user Can you give me a simple example?<|im_end|> <|im_start|>assistant
==========
--- REASONING_TASK --- Complete conversation format: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. Your role as an assistant involves thoroughly exploring questions through a systematic thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracking, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution using the specified format: Thought section Solution section. In the Thought section, detail your reasoning process in steps. Each step should include detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The Solution section should be logical, accurate, and concise and detail necessary steps needed to reach the conclusion. <|im_start|>user Solve step by step: If a train travels 120 miles in 2 hours, what is its average speed?<|im_end|> With generation prompt: <|im_start|>system ## Metadata Knowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. Your role as an assistant involves thoroughly exploring questions through a systematic thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracking, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution using the specified format: Thought section Solution section. In the Thought section, detail your reasoning process in steps. Each step should include detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The Solution section should be logical, accurate, and concise and detail necessary steps needed to reach the conclusion. <|im_start|>user Solve step by step: If a train travels 120 miles in 2 hours, what is its average speed?<|im_end|

> <|im_start|>assistant
==========

```
</details>

### Compare Base vs Instruct Model Responses

In this section, we run the same prompt through the base and instruct variants to
observe formatting differences and how the chat template impacts generation quality
and style.

```python
# Test the same prompt on both models
test_prompt = "Explain quantum computing in simple terms."

# Prepare the prompt for base model (no chat template)
base_inputs = base_tokenizer(test_prompt, return_tensors="pt").to(device)

# Prepare the prompt for instruct model (with chat template)
instruct_messages = []
instruct_formatted = instruct_tokenizer.apply_chat_template(
    instruct_messages,
    tokenize=False,
    add_generation_prompt=True
)
instruct_inputs = instruct_tokenizer(instruct_formatted,
return_tensors="pt").to(device)

# Generate responses
print("=== Model comparison ===\n")

print("🤖 BASE MODEL RESPONSE:")
with torch.no_grad():
    base_outputs = base_model.generate(
        **base_inputs,
        max_new_tokens=150,
        temperature=0.7,
        do_sample=True,
        pad_token_id=base_tokenizer.eos_token_id
    )
    base_response = base_tokenizer.decode(base_outputs[0],
skip_special_tokens=True)
    print(base_response[len(test_prompt):])  # Show only the generated part

print("\n" + "="*50)
print("Instruct model response:")
with torch.no_grad():
    instruct_outputs = instruct_model.generate(
        **instruct_inputs,
        max_new_tokens=150,
        temperature=0.7,
        do_sample=True,
        pad_token_id=instruct_tokenizer.eos_token_id
    )
```
```

```
    instruct_response = instruct_tokenizer.decode(instruct_outputs[0],
skip_special_tokens=True)
    # Extract only the assistant's response
    assistant_start = instruct_response.find("<|im_start|>assistant\n") + len("<|
im_start|>assistant\n")
    assistant_response = instruct_response[assistant_start:].split("<|im_end|>")[0]
    print(assistant_response)
```

If we dive into the out put below, we can see the differences between the base model and the instruct model. In short, the base model continues the string, while the instruct model uses the chat template. For example, the base model starts with `" What are the differences between the classical bit and the quantum bit?"`, while the instruct model starts by answering the question: `"Quantum computing is a type of computing that uses quantum bits"`.
Output
```python output === Model comparison === 🤖BASE MODEL RESPONSE: Why is it thought to be superior to our current technology? How is it superior? What is it's limit? Quantum computing is based on the fact that in quantum mechanics, a particle can be in multiple states at the same time. This is called superposition. But a single particle can not be in multiple locations at the same time. That is called entanglement. So, how can you have a particle in multiple locations at the same time? Quantum mechanics says that if you measure the location of a particle, it will randomly jump to a particular location. So, if you have 1000 particles, you can have each particle in 1000 different locations at the same time. This is very useful for solving problems. For example
==================================================
Instruct model response: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. user Explain quantum computing in simple terms. assistant Quantum computing is a type of computing that uses quantum bits, or qubits, to perform calculations. In traditional computers, we use bits, which can be either 0 or 1. But in quantum computing, we use qubits that can exist in multiple states at once, like both 0 and 1 simultaneously. Think of it like flipping a coin. A regular coin can land on either heads or tails, but a quantum coin can land on both heads and tails at the same time. This property is called superposition. Another unique aspect of quantum computing is entanglement. Imagine two coins that are linked together. If one coin lands on heads, the other coin will always land on tails, no matter how far apart they are

```
</details>

Test Dual-Mode Reasoning
```
```

````
Here we probe SmolLM3's reasoning mode with math and proportionality problems,
keeping temperature low for consistency and extracting only the assistant's
response from the chat-formatted output.

```python
# Test SmolLM3's reasoning capabilities
reasoning_prompts = [
    "What is 15 × 24? Show your work.",
    "A recipe calls for 2 cups of flour for 12 cookies. How much flour is needed
for 30 cookies?",
    "If I have $50 and spend $18.75 on lunch and $12.30 on a book, how much money
do I have left?"
]

print("=== TESTING REASONING CAPABILITIES ===\n")

for i, prompt in enumerate(reasoning_prompts, 1):
    print(f"Problem : ")

    messages = []
    formatted_prompt = instruct_tokenizer.apply_chat_template(
        messages, tokenize=False, add_generation_prompt=True
    )
    inputs = instruct_tokenizer(formatted_prompt, return_tensors="pt").to(device)

    with torch.no_grad():
        outputs = instruct_model.generate(
            **inputs,
            max_new_tokens=200,
            temperature=0.3,  # Lower temperature for more consistent reasoning
            do_sample=True,
            pad_token_id=instruct_tokenizer.eos_token_id
        )
        response = instruct_tokenizer.decode(outputs[0], skip_special_tokens=True)
        assistant_start = response.find("<|im_start|>assistant\n") + len("<|
im_start|>assistant\n")
        assistant_response = response[assistant_start:].split("<|im_end|>")[0]
        print(f"Answer: ")

    print("\n" + "-"*50 + "\n")
````

If we dive into the out put below, we can see that the instruct model's hybrid reasoning being applied with the `/no_think` mode. When the mode is activated, the model will enclose thinking process in `` tags. It uses these tokens to explore possible solutions and answer the question. After the thinking process, the model will provide the final answer, which we can extract with the chat template, or string manipulation here.

Output

```python output === TESTING REASONING CAPABILITIES === Thinking prompt: / no_think Problem 1: What is 15 × 24? Show your work. Answer: nowledge Cutoff

Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. user What is 15 × 24? Show your work. assistant To find the product of 15 and 24, we can use the standard multiplication algorithm. Here's how we can do it step by step:

```
   15
× 24

```

First, we multiply 15 by 4 (the units digit of 24):

```
   15
× 24
------
  60  (15 × 4)
```

Next, we multiply 15 by 20 (the tens digit of 24, shifted one place to the left):

```
   15
× 24
------
  60  (15 × 4)
 300 (15 × 20)
------
```

Now, we add the two partial products:

```
   15
× 24
------
  60  (15 × 4)
 300 (15 × 20)
------
 360
```

So, 15 × 24 = 360. ---------- Problem 2: A recipe calls for 2 cups of flour for 12 cookies. How much flour is needed for 30 cookies? Answer: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. user A recipe calls for 2 cups of flour for 12 cookies. How much flour is needed for 30 cookies? assistant To solve this problem, we need to determine the amount of flour needed per cookie and then multiply that by the number of cookies we want to make. First, let's find out how much flour is needed

per cookie. The recipe calls for 2 cups of flour for 12 cookies. To find the amount of flour per cookie, we divide the total amount of flour by the number of cookies: 2 cups / 12 cookies = 1/6 cup per cookie Now that we know how much flour is needed per cookie, we can multiply that by the number of cookies we want to make (30): 1/6 cup per cookie * 30 cookies = 5 cups So, to make 30 cookies, you would need 5 cups of flour. ---------- Problem 3: If I have $50 and spend $18.75 on lunch and $12.30 on a book, how much money do I have left? Answer: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /no_think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. user If I have $50 and spend $18.75 on lunch and $12.30 on a book, how much money do I have left? assistant To find out how much money you have left, you need to subtract the total amount spent from your initial amount. First, calculate the total amount spent on lunch and the book: $18.75 (lunch) + $12.30 (book) = $31.05 Now, subtract the total amount spent from your initial amount: $50 (initial amount) - $31.05 (total spent) = $18.95 So, you have $18.95 left. ---------- Thinking prompt: /think Problem 1: What is 15 × 24? Show your work. Answer: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. Your role as an assistant involves thoroughly exploring questions through a systematic thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracking, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution using the specified format: Thought section Solution section. In the Thought section, detail your reasoning process in steps. Each step should include detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The Solution section should be logical, accurate, and concise and detail necessary steps needed to reach the conclusion. user What is 15 × 24? Show your work. assistant Okay, let's see. I need to calculate 15 multiplied by 24. Hmm, how do I do that? I remember there are a few methods. Maybe the standard multiplication algorithm? Or maybe breaking it down into smaller parts. Let me try both ways to make sure I get the right answer. First, the standard way. Let me write it out like I'm doing long multiplication. So, 15 times 24. I can think of 24 as 20 + 4. So maybe I can break it down into 15 times 20 plus 15 times 4. That might be easier. Starting with 15 times 20. Well, 15 times 2 is 30, so adding a zero at the end makes

it 300. So 15 × 20 = 300. Got that part. Now, 15 times 4. Let me calculate that. 15 times 4 is 60. Right? Because 10 ---------- Problem 2: A recipe calls for 2 cups of flour for 12 cookies. How much flour is needed for 30 cookies? Answer: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. Your role as an assistant involves thoroughly exploring questions through a systematic thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracking, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution using the specified format: Thought section Solution section. In the Thought section, detail your reasoning process in steps. Each step should include detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The Solution section should be logical, accurate, and concise and detail necessary steps needed to reach the conclusion. user A recipe calls for 2 cups of flour for 12 cookies. How much flour is needed for 30 cookies? assistant Okay, so I need to figure out how much flour is needed for 30 cookies if the recipe calls for 2 cups of flour for 12 cookies. Hmm, let's see. I think this is a proportion problem. If 12 cookies require 2 cups, then I need to find out how much flour is needed for 30 cookies. First, maybe I should determine how much flour is needed per cookie. If 12 cookies take 2 cups, then per cookie it would be 2 divided by 12. Let me write that down: 2 cups / 12 cookies. That simplifies to 1/6 cup per cookie. Wait, 2 divided by 12 is 1/6? Let me check that again. 2 divided by 12 is indeed 1/6. Yeah, because 12 divided by 6 is 2, so 2 divided by 12 is 1/6. So each cookie needs ---------- Problem 3: If I have $50 and spend $18.75 on lunch and $12.30 on a book, how much money do I have left? Answer: nowledge Cutoff Date: June 2025 Today Date: 03 September 2025 Reasoning Mode: /think ## Custom Instructions You are a helpful AI assistant named SmolLM, trained by Hugging Face. Your role as an assistant involves thoroughly exploring questions through a systematic thinking process before providing the final precise and accurate solutions. This requires engaging in a comprehensive cycle of analysis, summarizing, exploration, reassessment, reflection, backtracking, and iteration to develop well-considered thinking process. Please structure your response into two main sections: Thought and Solution using the specified format: Thought section Solution section. In the Thought section, detail your reasoning process in steps. Each step should include

detailed considerations such as analysing questions, summarizing relevant findings, brainstorming new ideas, verifying the accuracy of the current steps, refining any errors, and revisiting previous steps. In the Solution section, based on various attempts, explorations, and reflections from the Thought section, systematically present the final solution that you deem correct. The Solution section should be logical, accurate, and concise and detail necessary steps needed to reach the conclusion. user If I have $50 and spend $18.75 on lunch and $12.30 on a book, how much money do I have left? assistant Okay, let's see. The problem is about calculating how much money is left after spending on lunch and a book. I need to start with the initial amount of $50 and then subtract the amounts spent on lunch and the book. First, I should add up the total amount spent. The lunch cost $18.75 and the book cost $12.30. To add these two amounts together, I need to make sure they are in the same units, which they already are since both are in dollars. So, adding $18.75 and $12.30. Let me do that step by step. Starting with the dollars: 18 dollars plus 12 dollars is 30 dollars. Then the cents: 75 cents plus 30 cents is 105 cents. Now, 105 cents is equal to $1.05 because 100 cents make a dollar, so 105 cents is 1 dollar and 5 cents. Therefore, adding the dollars and cents together
---

```
</details>

### Validation

Run the code above and verify that you can see:
1. Different chat template formats for various conversation types
2. Clear differences between base model and instruct model responses
3. SmolLM3's reasoning capabilities in action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## Exercise 2: Dataset Processing for SFT

**Objective**: Learn to process and prepare datasets for supervised fine-tuning
using SmolTalk2 and other datasets.

### Explore the SmolTalk2 Dataset

We load the SmolTalk2 SFT split, inspect its structure and a few samples to
understand fields (e.g., `messages`) and available subsets before preparing data
for training.

```python
# Load and explore the SmolTalk2 dataset
print("=== EXPLORING SMOLTALK2 DATASET ===\n")

# Load the SFT subset
```
```

```
dataset_dict = load_dataset("HuggingFaceTB/smoltalk2", "SFT")
print(f"Total splits: ")
print(f"Available splits: ")
print(f"Number of total rows: ")
print(f"Dataset structure: ")
```

If we dive into the out put below, we can see the structure of the dataset. It has 25 splits, and the total number of rows is 3,383,242.

Output

```python output === EXPLORING SMOLTALK2 DATASET === Resolving data files: 100% 124/124 [00:00<00:00, 9963.48it/s] Resolving data files: 100% 113/113 [00:00<00:00, 57.54it/s] Resolving data files: 100% 113/113 [00:00<00:00, 114.07it/s] Loading dataset shards: 100% 105/105 [00:00<00:00, 2570.62it/s] Total splits: 25 Available splits: ['LongAlign_64k_Qwen3_32B_yarn_131k_think', 'OpenThoughts3_1.2M_think', 'aya_dataset_Qwen3_32B_think', 'multi_turn_reasoning_if_think', 's1k_1.1_think', 'smolagents_toolcalling_traces_think', 'smoltalk_everyday_convs_reasoning_Qwen3_32B_think', 'smoltalk_multilingual8_Qwen3_32B_think', 'smoltalk_systemchats_Qwen3_32B_think', 'table_gpt_Qwen3_32B_think', 'LongAlign_64k_context_lang_annotated_lang_6_no_think', 'Mixture_of_Thoughts_science_no_think', 'OpenHermes_2.5_no_think', 'OpenThoughts3_1.2M_no_think_no_think', 'hermes_function_calling_v1_no_think', 'smoltalk_multilingual_8languages_lang_5_no_think', 'smoltalk_smollm3_everyday_conversations_no_think', 'smoltalk_smollm3_explore_instruct_rewriting_no_think', 'smoltalk_smollm3_smol_magpie_ultra_no_think', 'smoltalk_smollm3_smol_rewrite_no_think', 'smoltalk_smollm3_smol_summarize_no_think', 'smoltalk_smollm3_systemchats_30k_no_think', 'table_gpt_no_think', 'tulu_3_sft_personas_instruction_following_no_think', 'xlam_traces_no_think'] Number of total rows: 3383242 Dataset structure: DatasetDict() OpenThoughts3_1.2M_think: Dataset() aya_dataset_Qwen3_32B_think: Dataset() multi_turn_reasoning_if_think: Dataset() s1k_1.1_think: Dataset() smolagents_toolcalling_traces_think: Dataset() smoltalk_everyday_convs_reasoning_Qwen3_32B_think: Dataset() smoltalk_multilingual8_Qwen3_32B_think: Dataset() smoltalk_systemchats_Qwen3_32B_think: Dataset() table_gpt_Qwen3_32B_think: Dataset() LongAlign_64k_context_lang_annotated_lang_6_no_think: Dataset() Mixture_of_Thoughts_science_no_think: Dataset() OpenHermes_2.5_no_think:
```

Dataset() OpenThoughts3_1.2M_no_think_no_think: Dataset()
hermes_function_calling_v1_no_think: Dataset()
smoltalk_multilingual_8languages_lang_5_no_think: Dataset()
smoltalk_smollm3_everyday_conversations_no_think: Dataset()
smoltalk_smollm3_explore_instruct_rewriting_no_think: Dataset()
smoltalk_smollm3_smol_magpie_ultra_no_think: Dataset()
smoltalk_smollm3_smol_rewrite_no_think: Dataset()
smoltalk_smollm3_smol_summarize_no_think: Dataset()
smoltalk_smollm3_systemchats_30k_no_think: Dataset() table_gpt_no_think:
Dataset() tulu_3_sft_personas_instruction_following_no_think: Dataset()
xlam_traces_no_think: Dataset() })

````
</details>

### Process Different Dataset Types

The SmolTalk2 dataset is a collection of open source datasets compiled together for
convenience. It contains a mixture of useful post training use cases, like tool
use, long context, and more. They are all in chat format, which is easy to use for
training. However, not all datasets are shared in consistent format so often we
need to process them into a unified chat `messages` layout.

For this exercise, we will standardize multiple dataset formats into a unified chat
`messages` layout. We define lightweight processors for QA and instruction datasets
and walk through a concrete example using GSM8K.

```python
# Function to process different dataset formats
def process_qa_dataset(examples, question_col, answer_col):
    """Process Q&A datasets into chat format"""
    processed = []

    for question, answer in zip(examples[question_col], examples[answer_col]):
        messages = [
            ,

        ]
        processed.append(messages)

    return

def process_instruction_dataset(examples):
    """Process instruction-following datasets"""
    processed = []

    for instruction, response in zip(examples["instruction"],
examples["response"]):
        messages = [
            ,
````
````

```
        ]
        processed.append(messages)

    return

# Example: Process GSM8K math dataset
print("=== PROCESSING GSM8K DATASET ===\n")

gsm8k = load_dataset("openai/gsm8k", "main", split="train[:100]")  # Small subset
for demo
print(f"Original GSM8K example: ")

# Convert to chat format
def process_gsm8k(examples):
    processed = []
    for question, answer in zip(examples["question"], examples["answer"]):
        messages = [
            ,
            ,

        ]
        processed.append(messages)
    return

gsm8k_processed = gsm8k.map(process_gsm8k, batched=True,
remove_columns=gsm8k.column_names)
print(f"Processed example: ")
```

Below we find two samples from the two separate datasets in the same format.
Output
```python output === PROCESSING GSM8K DATASET === README.md:
7.94k/?  [00:00<00:00,  572kB/s] main/train-00000-of-00001.parquet:  100%
2.31M/2.31M  [00:01<00:00,  42.6kB/s] main/test-00000-of-00001.parquet:  100%
419k/419k  [00:00<00:00,  813kB/s] Generating  train  split:  100%
7473/7473  [00:00<00:00,  321312.49  examples/s] Generating  test  split:  100%
1319/1319  [00:00<00:00,  97120.71  examples/s] Original GSM8K example:
Map:  100%  100/100  [00:00<00:00,  4792.50  examples/s] Processed
example: , , ]}

```
</details>

Apply Chat Templates to Datasets

Once messages are normalized, we apply the model's chat template to convert each example into plain training text (`text` column) suitable for language modeling with SFT.

```python
# Function to apply chat templates to processed datasets
def apply_chat_template_to_dataset(dataset, tokenizer):
```

```
    """Apply chat template to dataset for training"""

    def format_messages(examples):
        formatted_texts = []

        for messages in examples["messages"]:
            # Apply chat template
            formatted_text = tokenizer.apply_chat_template(
                messages,
                tokenize=False,
                add_generation_prompt=False  # We want the complete conversation
            )
            formatted_texts.append(formatted_text)

        return

    return dataset.map(format_messages, batched=True)

# Apply to our processed GSM8K dataset
gsm8k_formatted = apply_chat_template_to_dataset(gsm8k_processed,
instruct_tokenizer)
print("=== FORMATTED TRAINING DATA ===")
print(gsm8k_formatted[0]["text"])
```

Output

```python output === PROCESSING GSM8K DATASET === README.md: 7.94k/? [00:00<00:00, 572kB/s] main/train-00000-of-00001.parquet: 100% 2.31M/2.31M [00:01<00:00, 42.6kB/s] main/test-00000-of-00001.parquet: 100% 419k/419k [00:00<00:00, 813kB/s] Generating train split: 100% 7473/7473 [00:00<00:00, 321312.49 examples/s] Generating test split: 100% 1319/1319 [00:00<00:00, 97120.71 examples/s] Original GSM8K example: Map: 100% 100/100 [00:00<00:00, 4792.50 examples/s] Processed example: , , ]}

```
</details>

Exercise 3: Fine-Tuning SmolLM3 with SFTTrainer

Objective: Perform supervised fine-tuning on SmolLM3 using TRL's SFTTrainer
with real datasets.

> [!WARNING]
> You will need a GPU with at least 8GB VRAM.

Step 1: Setup and Model Loading

We load the base model and tokenizer, set padding behavior, and move the model to
an appropriate device to prepare for fine-tuning.
```
```

```python
# Import required libraries for fine-tuning
from transformers import AutoModelForCausalLM, AutoTokenizer, TrainingArguments
from trl import SFTTrainer, SFTConfig, DataCollatorForCompletionOnlyLM
from datasets import load_dataset



# Initialize Weights & Biases (optional)
# wandb.init(project="smollm3-finetuning")

# Load SmolLM3 base model for fine-tuning
model_name = "HuggingFaceTB/SmolLM3-3B-Base"
new_model_name = "SmolLM3-Custom-SFT"

print(f"Loading ...")
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    dtype=torch.bfloat16,
    device_map="auto",
    trust_remote_code=True
)

tokenizer = AutoTokenizer.from_pretrained(model_name)
tokenizer.pad_token = tokenizer.eos_token  # Set padding token
tokenizer.padding_side = "right"  # Padding on the right for generation

print(f"Model loaded! Parameters: ")
```

```python output Loading HuggingFaceTB/SmolLM3-3B-Base... Model loaded! Parameters: 3,075,098,624

```
Dataset Preparation

Here we select a manageable subset for speed, then map each example to a single `text` string by applying the chat template—this is the field the trainer will read.

```python
# Load and prepare training dataset
print("=== PREPARING DATASET ===\n")

# Option 1: Use SmolTalk2 (recommended for beginners)
dataset = load_dataset("HuggingFaceTB/smoltalk2", "SFT")
train_dataset =
dataset["smoltalk_everyday_convs_reasoning_Qwen3_32B_think"].select(range(1000))  #
Use subset for faster training

# Option 2: Use your own processed dataset from Exercise 2
# train_dataset = gsm8k_formatted.select(range(500))

print(f"Training examples: ")
print(f"Example: ")
```

```
# Prepare the dataset for SFT
def format_chat_template(example):
    """Format the messages using the chat template"""
    if "messages" in example:
        # SmolTalk2 format
        messages = example["messages"]
    else:
        # Custom format - adapt as needed
        messages = [
            ,

        ]

    # Apply chat template
    text = instruct_tokenizer.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=False
    )
    return

# Apply formatting
formatted_dataset = train_dataset.map(format_chat_template)
formatted_dataset = formatted_dataset.remove_columns(
    [col for col in formatted_dataset.column_names if col != "text"]
)
print(f"Formatted example: ...")
```

Output
```python output === PREPARING DATASET === Training examples: 1000 Example: , ]} Formatted example: You are a math tutor. Solve problems step by step. Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?...

```
</details>

Training Configuration

We configure key knobs for SFT (batch size, sequence length, learning rate, logging/saving cadence) and enable optional tracking and Hub integration.

```python
# Configure training parameters
training_config = SFTConfig(
    # Model and data
    output_dir=f"./",
    dataset_text_field="text",
    max_length=2048,

    # Training hyperparameters
    per_device_train_batch_size=2,  # Adjust based on your GPU memory
```

```
    gradient_accumulation_steps=2,
    learning_rate=5e-5,
    num_train_epochs=1,  # Start with 1 epoch
    max_steps=500,  # Limit steps for demo

    # Optimization
    warmup_steps=50,
    weight_decay=0.01,
    optim="adamw_torch",

    # Logging and saving
    logging_steps=10,
    save_steps=100,
    eval_steps=100,
    save_total_limit=2,

    # Memory optimization
    dataloader_num_workers=0,
    group_by_length=True,  # Group similar length sequences

    # Hugging Face Hub integration
    push_to_hub=False,  # Set to True to upload to Hub
    hub_model_id=f"your-username/",

    # Experiment tracking
    report_to=["trackio"],  # Use trackio for experiment tracking
    run_name=f"-training",
)

print("Training configuration set!")
print(f"Effective batch size: ")
```

```python output Training configuration set! Effective batch size: 4
```

```
### Optional: Train with LoRA/PEFT (memory-efficient)

If you have limited GPU memory or want faster iterations, use LoRA via PEFT. This
trains only small adapter weights while keeping the base model frozen, then you can
either keep using adapters or merge them later for deployment.

```python
# LoRA configuration with PEFT
from peft import LoraConfig

peft_config = LoraConfig(
    r=8,
    lora_alpha=16,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
)

# Create SFTTrainer with LoRA enabled
from trl import SFTTrainer
```
```

```
lora_trainer = SFTTrainer(
    model=model,
    train_dataset=formatted_dataset,  # dataset with a "text" field or messages +
dataset_text_field in config
    args=training_config,
    peft_config=peft_config,  # << enable LoRA
)

print("Starting LoRA training…")
lora_trainer.train()
```

Step 4: Initialize SFTTrainer and Train We instantiate the trainer, capture a pre-training baseline generation, launch `train()`, and save the resulting checkpoints to the configured output directory.

```
trainer = SFTTrainer(
    model=model,
    train_dataset=formatted_dataset,
    args=config,
)
```

And we can train the model.

```
trainer.train()
```

Test the Fine-Tuned Model Finally, we regenerate the same prompt to qualitatively compare outputs before vs after training, and optionally push the model to the Hub for sharing.

```
# Test the fine-tuned model
print("=== AFTER TRAINING ===")
with torch.no_grad():
    outputs = model.generate(
        **inputs,
        max_new_tokens=100,
        temperature=0.7,
        do_sample=True,
        pad_token_id=tokenizer.eos_token_id
    )
    response = tokenizer.decode(outputs[0], skip_special_tokens=True)
    print(response[len(formatted_prompt):])

# Optional: Push to Hugging Face Hub
if training_config.push_to_hub:
    trainer.push_to_hub(
        commit_message="Fine-tuned SmolLM3 with custom dataset",
        tags=["smol-course", "sft", "instruction-tuning"]
```

```
    )
    print(f"Model pushed to Hub: ")
```

--- ## Exercise 4: Production Workflow with TRL CLI In the previous exercises we've dived deep into using TRL's Python API for fine-tuning and explored the data we're using and generating. In this exercise we'll explore using the TRL CLI to fine-tune a model. This will be the most common way to fine-tune a model in production. We can define a command in TRL CLI to fine-tune a model. We'll be able to run it with `trl sft` command. The CLI command and Python API share the same configuration options. We preprocessed the `smoltalk_everyday_convs_reasoning_Qwen3_32B_think` subset of SmolTalk2 so that is easier to work with it when using the TRL CLI.

```
# Fine-tune SmolLM3 using TRL CLI
trl sft \
    --model_name_or_path HuggingFaceTB/SmolLM3-3B-Base \
    --dataset_name HuggingFaceTB/smoltalk2_everyday_convs_think \
    --output_dir ./smollm3-sft-cli \
    --per_device_train_batch_size 4 \
    --gradient_accumulation_steps 2 \
    --learning_rate 5e-5 \
    --num_train_epochs 1 \
    --max_length 2048 \
    --logging_steps 10 \
    --save_steps 500 \
    --warmup_steps 100 \
    --bf16 True \
    --push_to_hub \
    --hub_model_id your-username/smollm3-sft-cli
```

For convenience and reproducibility, we can also create a configuration file to fine-tune a model. For example, we could create a file called `sft_config.yaml` and put the following content in it:

```
# Model and dataset
model_name_or_path: HuggingFaceTB/SmolLM3-3B-Base
dataset_name: HuggingFaceTB/smoltalk2_everyday_convs_think
output_dir: ./smollm3-advanced-sft

# Training hyperparameters
per_device_train_batch_size: 2
gradient_accumulation_steps: 4
learning_rate: 3e-5
num_train_epochs: 2
max_length: 4096

# Optimization
warmup_steps: 200
```

```
weight_decay: 0.01
optim: adamw_torch
lr_scheduler_type: cosine

# Memory and performance
bf16: true
dataloader_num_workers: 4
group_by_length: true
remove_unused_columns: false

# Logging and evaluation
logging_steps: 25
eval_steps: 250
save_steps: 500
eval_strategy: steps
load_best_model_at_end: true
metric_for_best_model: eval_loss

# Hub integration
push_to_hub: true
hub_model_id: your-username/smollm3-advanced
hub_strategy: every_save
```

We could then commit this file to the repository and track it with Git.

```
# Run training with config file
trl sft --config sft_config.yaml
```

Troubleshooting **If you get GPU out of memory errors:** - Reduce `per_device_train_batch_size` to 1 - Reduce `max_length` to 1024 or 512 - Use `torch.cuda.empty_cache()` to clear GPU memory **If models fail to load:** - Check your internet connection - Try using `device_map="cpu"` for CPU loading - Use a smaller model like `HuggingFaceTB/SmolLM3-1.7B` for testing **If training fails:** - Make sure your dataset is properly formatted - Check that all examples have reasonable length (not too long) - Monitor the training loss - it should decrease steadily ## Conclusion Congratulations! You've completed comprehensive hands-on exercises covering: - SmolLM3's chat template system and dual-mode reasoning - Dataset processing and preparation techniques - Supervised fine-tuning with Python APIs - Production workflows using CLI tools - Distributed training setups These skills form the foundation for building sophisticated instruction-tuned models. In the next modules, we'll explore preference alignment, parameter-efficient fine-tuning, and advanced evaluation techniques. ## Resources for Further Learning - [TRL Documentation](https://huggingface.co/docs/trl) - Complete reference - [SmolLM3 Model Card](https://huggingface.co/HuggingFaceTB/SmolLM3-3B) - Model details - [SmolTalk2 Dataset](https://huggingface.co/datasets/HuggingFaceTB/smoltalk2) -

Training data - [Hugging Face Hub](https://huggingface.co/models) - Share your models - [Discord Community](https://discord.gg/UrrTSsSyjb) - Get help and discuss

Training with Hugging Face Jobs

Hugging Face Jobs provides fully managed cloud infrastructure for training models without the hassle of setting up GPUs, managing dependencies, or configuring environments locally. This is particularly valuable for SFT training, which can be resource-intensive and time-consuming.

Why Use Jobs for SFT Training?

- **Scalable Infrastructure**: Access to high-end GPUs (A100, L4, etc.) without hardware investment
- **Zero Setup**: No need to manage CUDA drivers, Docker containers, or environment configurations
- **Cost Effective**: Pay only for compute time used, with automatic shutdown after completion
- **Integrated Workflow**: Seamless integration with Hugging Face Hub for model storage and sharing
- **Monitoring**: Built-in logging and progress tracking through the Hub interface

Requirements

To use Hugging Face Jobs, you need: - A **Pro, Team, or Enterprise** Hugging Face plan which you can get here - Authentication via `hf auth login`

Running SFT with Jobs: Two Approaches

The best way to run TRL with HF jobs is using the built-in scripts. They take advantage of `uv` to manage dependencies and `hf jobs` to run the training job.

This guide will walk you through using TRL's built-in scripts to train a model with Hugging Face Jobs. If you want to use a custom script, you can implement `uv` dependencies and run the script with `hf jobs run`.

Create a custom training script with inline dependencies

```
# sft_training.py
# /// script
# dependencies = [
#     "trl[sft]>=0.7.0",
#     "transformers>=4.36.0",
#     "datasets>=2.14.0",
#     "accelerate>=0.24.0",
#     "peft>=0.7.0"
# ]
# ///

from trl import SFTTrainer, SFTConfig
from transformers import AutoModelForCausalLM, AutoTokenizer
from datasets import load_dataset

# Load model and tokenizer
model = AutoModelForCausalLM.from_pretrained("HuggingFaceTB/SmolLM3-3B-Base")
tokenizer = AutoTokenizer.from_pretrained("HuggingFaceTB/SmolLM3-3B-Base")

# Load dataset
dataset = load_dataset("HuggingFaceTB/smoltalk2", "SFT")

# Configure training
config = SFTConfig(
    output_dir="./smollm3-jobs-sft",
    per_device_train_batch_size=4,
    learning_rate=5e-5,
    max_steps=1000,
    logging_steps=50,
    save_steps=200,
    push_to_hub=True,
    hub_model_id="your-username/smollm3-jobs-sft"
)

# Train
trainer = SFTTrainer(
    model=model,
    train_dataset=dataset["smoltalk_everyday_convs_reasoning_Qwen3_32B_think"],
    args=config,
)
trainer.train()
```

Then run with the Jobs CLI:

```
# Run the UV script on Jobs
hf jobs uv run \
    --flavor a10g-large \
    --timeout 2h \
    --secrets HF_TOKEN \
    sft_training.py
```

Hardware Selection for SFT

Choose the right hardware flavor based on your model size and training requirements:

For SmolLM3-3B (Recommended): - `a10g-large`: 24GB GPU memory, cost-effective for most SFT tasks - `a100-large`: 40GB GPU memory, fastest training with larger batch sizes - `l4x1`: 24GB GPU memory, multi-GPU setup for distributed training

For Larger Models (7B+): - `a100-large`: Required for 7B+ models - `l4x4`: Multi-GPU setup for distributed training

Budget Options: - `t4-small`: 16GB GPU memory, slower but economical for experimentation - `l4x1`: 24GB GPU memory, good balance of cost and performance

> [!TIP] For a detailed comparison of the different hardware flavors, you can check out the Pricing Page page.

Advanced Jobs Configuration

```
# Use TRL's maintained SFT script directly
hf jobs uv run \
    --flavor a10g-large \
    --timeout 2h \
    --secrets HF_TOKEN \
    "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/sft.py" \
    --model_name_or_path HuggingFaceTB/SmolLM3-3B-Base \
    --dataset_name HuggingFaceTB/smoltalk2_everyday_convs_think \
    --learning_rate 5e-5 \
    --per_device_train_batch_size 4 \
    --max_steps 1000 \
    --output_dir smollm3-sft-jobs \
    --push_to_hub \
    --hub_model_id your-username/smollm3-sft \
    --report_to trackio
```

Environment Variables and Secrets:

If you're working with a custom script, you can use the `--secrets` flag to pass in environment variables.

```
hf jobs uv run \
    --flavor a10g-large \
```

```
    --timeout 3h \
    --secrets HF_TOKEN=your_token \
    --secrets WANDB_API_KEY=your_wandb_key \
    --env WANDB_PROJECT=smollm3-sft \
    --env CUDA_VISIBLE_DEVICES=0 \
    my_sft_training.py
```

Monitoring Your Training Job

To check you training job, you can use the `hf jobs` command or you can go to Job Settings on the Hub.

Check Job Status:

```
# List all jobs
hf jobs ps -a

# Get detailed job information
hf jobs inspect <job_id>

# Stream job logs in real-time
hf jobs logs <job_id> --follow

# Cancel a running job if needed
hf jobs cancel <job_id>
```

LoRA/PEFT on Jobs (optional)

Enable LoRA when using TRL's maintained SFT script by passing PEFT flags. See the script for authoritative flags and defaults: `https://github.com/huggingface/trl/blob/main/trl/scripts/sft.py`.

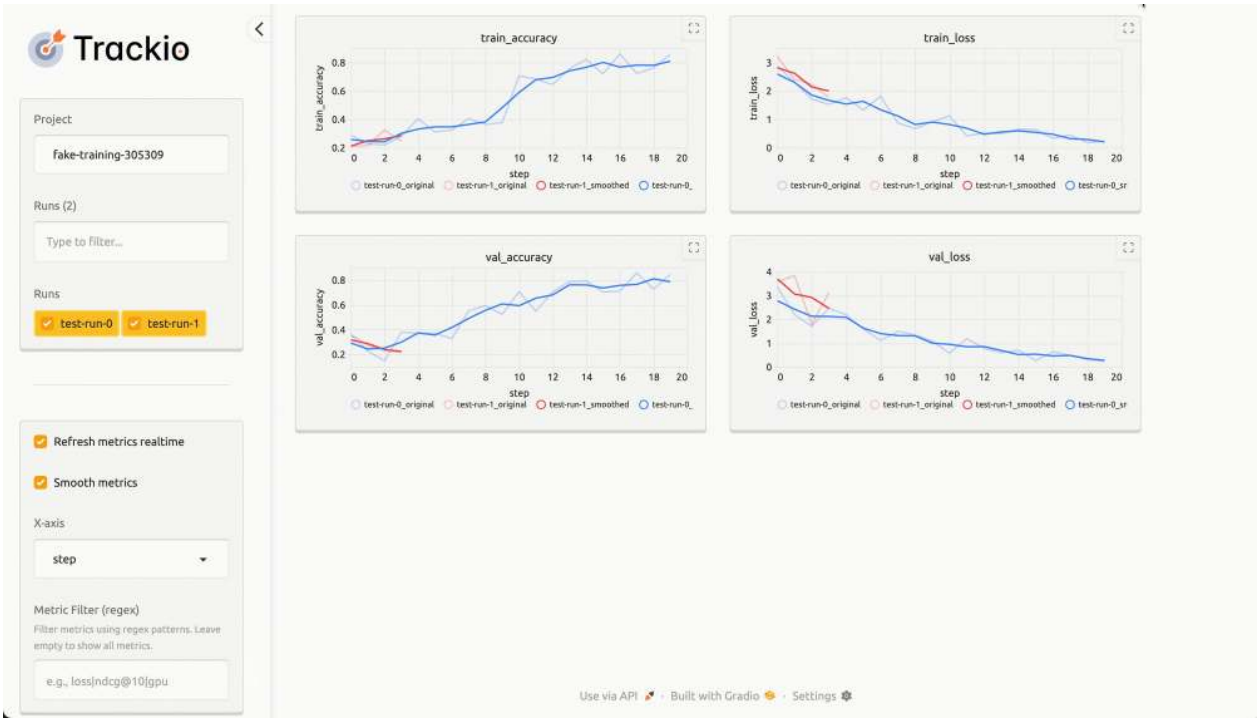
```
hf jobs uv run \
  --flavor a10g-large \
  --timeout 2h \
  --secrets HF_TOKEN \
  "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/sft.py" \
  --model_name_or_path HuggingFaceTB/SmolLM3-3B-Base \
  --dataset_name HuggingFaceTB/smoltalk2_everyday_convs_think \
  --output_dir smollm3-lora-sft-jobs \
  --per_device_train_batch_size 4 \
  --learning_rate 5e-5 \
  --max_steps 1000 \
  --report_to trackio \
  --push_to_hub \
  --hub_model_id your-username/smollm3-lora-sft \
  --use_peft \
  --lora_r 8 \
```

```
  --lora_alpha 16 \
  --lora_dropout 0.05 \
  --lora_target_modules all-linear
```

Notes: - Confirm flag names in the TRL SFT script before running: `https://github.com/huggingface/trl/blob/main/trl/scripts/sft.py`. - LoRA trains small adapters, which you can keep separate or merge later for deployment.

Monitoring with Trackio

You can monitor your training job with Trackio.



Cost Estimation

Approximate costs for SmolLM3-3B SFT training (1000 steps): - l4x1: ~$3-4 per hour (24GB GPU memory) - a10g-large: ~$4-6 per hour (24GB GPU memory) - a100-large: ~$8-12 per hour (40GB GPU memory)

Training typically takes 30-90 minutes for 1000 steps depending on hardware and configuration, making Jobs cost-effective compared to local GPU rental or cloud instances.

Cost-Saving Tips: - Use smaller batch sizes with gradient accumulation to fit on cheaper GPUs - Start with shorter training runs (500 steps) to validate your setup -

Use `l4x1` for initial experiments, then scale to faster GPUs for production - Set appropriate timeouts to avoid unexpected charges

Troubleshooting Common Issues

Out of Memory Errors: - Reduce `per_device_train_batch_size` - Enable gradient checkpointing - Use smaller `max_length`

Timeout Issues: - Increase timeout parameter - Reduce training steps or use more powerful hardware - Optimize data loading and preprocessing

Authentication Errors: - Ensure HF_TOKEN is correctly set as a secret - Verify your Hugging Face account has the required plan - Check token permissions for model uploads

Resources and Further Reading

- Hugging Face Jobs Documentation - Complete Jobs guide
- TRL Jobs Training Guide - TRL-specific Jobs examples
- Jobs Pricing - Current pricing for different hardware flavors
- Jobs CLI Reference - Command-line interface details

Submit your final project!

It's time to submit your project! This year the smol course will use a leaderboard based submission. Here's the plan:

1. Read the written guide for the chapter ✅
2. Train a model using what you learned in the chapter.
3. Push the model to the Hugging Face Hub.
4. Evaluate the model using `hf jobs`.
5. Open a pull request on the leaderboard.

On this page we will go through each step.

1. Read the written guide for the chapter and 2. Train a model using what you learned in the chapter.

For chapter 1's submission, you should read all the materials in the chapter and train a model using what you learned. Most of the training code is in the page on Supervised Fine-Tuning, but you'll need to combine this with the code on Chat Templates and the code on Training with Hugging Face Jobs.

3. Push the model to the Hugging Face Hub

Once you've trained a model, you'll need to push it to a repo on the Hugging Face Hub. In fact, TRL will take care of this for you if you add the `--push_to_hub` flag to your training command. So let's say you trained a model using `hf jobs`, then this parameter will look like this:

```
hf jobs uv run \
    --flavor a100-large \
    --secrets HF_TOKEN \
    "trl/scripts/sft.py" \
    ...
    --push_to_hub # this will push the model to the Hugging Face Hub
```

Your trained model will be available at `your-username/your-model-name`. For detailed documentation, check out the checkpoints documentation from `transformers`.

4. Evaluate the model using `hf jobs`

Now, we will need to evaluate the model. We will use `hf jobs` to evaluate the model as well and combine it with `openbench`. We will push the evaluation results to a dataset on the hub.

```
hf jobs uv run \ # run a hf jobs job with uv
--flavor a10g-large \ # select the machine size
--with "lighteval[vllm]" \ # install lighteval with vllm dependencies
s HF_TOKEN \ # share the huggingface write token
lighteval vllm "model_name=<your-username>/<your-model-name>" "lighteval|gsm8k|0|0"
--push-to-hub --results-org <your-username>
```

This command will evaluate the model using `lighteval` and `vllm` and save the results to the Hugging Face Hub in the dataset repo that you defined.

> [!TIP] We have not explored evaluation in this course yet, but in chapter 2 we will explore evaluation in more detail. For now, we're focusing on training and submitting your model.

5. Open a pull request on the leaderboard space

You are now ready to submit your model to the leaderboard! You need to do two things:

1. add your model's results to `submissions.json`
2. share you evaluation command (using `hf jobs`) in the PR text.

Add your model's results to `submissions.json`

Open a pull request on the leaderboard space to submit your model. You just need to model info and reference to the dataset you created in the previous step. We will pull the results and display them on the leaderboard.

```
    ]
}
```

Share you evaluation command in the PR text.

Within the PR text, share you evaluation command. For example:

```
hf jobs uv run ...
```

This will help us to reproduce your model evaluation before we add it to the leaderboard.

Wait for the PR to be merged

Once the PR is merged, your model will be added to the leaderboard! You can check the leaderboard here.

Test your knowledge

You've completed the unit — great work! Now put your learning to the test by taking the quiz.

Introduction to Vision Language Models

Vision Language Models (VLMs) can understand both images and text simultaneously, enabling tasks like image captioning, visual question answering, and multimodal reasoning. Just like LLMs, VLMs are trained to predict the next token — but with the added ability to process visual information. For example, `HuggingFaceTB/SmolVLM2-2.2B-Base` is a base VLM model, while `HuggingFaceTB/SmolVLM2-2.2B-Instruct` is instruction-tuned for chat-like interactions with users.

In this unit, we will explore how these models are built, how they work, and, most importantly, how you can use and adapt them for your own projects.

> [!TIP] By the end of this unit, you'll fine-tune a VLM using the same techniques you've already learned in previous units (like SFT). As ever, this unit is *smol but fast*!
>
> If you're looking for a deeper dive into computer vision, check out The Community Computer Vision Course.
>
> After completing this unit (and the assignment), don't forget to test your knowledge with the quiz!

What are Vision Language Models?

VLMs process image alongside text to enable tasks like image captioning, visual question answering, and multimodal reasoning.

A typical VLM architecture consists of an image encoder to extract visual features, a projection layer to align visual and textual representations, and a language model to process or generate text. This allows the model to establish connections between visual elements and language concepts.

VLMs can be used in different configurations depending on the use case. Base models handle general vision-language tasks, while chat-optimized variants support conversational interactions. Some models include additional components for grounding predictions in visual evidence or specializing in specific tasks like object detection.

Latest trends

Adding vision to language models has unlocked many exciting directions, including:

- **Reasoning-focused VLMs:** solve complex problems using visual inputs.
- **Specialized VLMs:** e.g. object detection, segmentation, or document understanding.
- **Vision-Language-Action models:** generate end actions for robotics.
- **Agentic VLMs:** enable complex workflows like chatting with documents or interacting with computer through screenshots.
- **Any-to-any models:** expanding beyond vision and text to handle multiple input/output modalities (text, image, audio, video, etc.).

Adapting Vision Language Models for specific needs

Fine-tuning a VLM means adapting a pre-trained model to your dataset or task. You've already seen strategies like supervised fine-tuning (SFT) and preference alignment in previous units, the same ideas apply here.

While the core tools and techniques remain similar to those used for LLMs, fine-tuning VLMs brings additional challenges. A key one is **data representation**: images must be carefully prepared so the model can effectively combine visual and textual information. Another factor is **model size**. VLMs are often much larger than LLMs, making efficiency critical.

To keep training practical and cost-effective, we can rely on techniques like **quantization** and **PEFT (Parameter-Efficient Fine-Tuning)**, as we explored in Unit 1. These approaches make fine-tuning more lightweight, enabling more users to adapt and experiment with powerful VLMs.

Evaluating Vision Language Models

As we saw in Unit 2, evaluation is a crucial step both during development and at production stage. For Vision Language Models (VLMs), the same principle applies: we need benchmarks to assess their capabilities and limitations during development, and real-world testing to ensure reliability and practical usefulness once deployed.

Some widely used **general-purpose benchmarks** include:

- **MMMU & MMMU-Pro:** large multi-discipline benchmarks requiring reasoning across domains like arts, science, and engineering.
- **MMBench:** over 3,000 single-choice questions testing skills such as OCR, localization, and reasoning.
- **MMT-Bench:** focuses on expert-level multimodal tasks, including recognition, localization, reasoning, and planning.

There are also **domain-specific benchmarks** designed to test specialized skills:

- **MathVista:** evaluates mathematical reasoning in the context of images.
- **AI2D:** focuses on diagram understanding.
- **ScienceQA:** science question answering.
- **OCRBench:** assesses document understanding and OCR capabilities.

Finally, for a **streamlined evaluation workflow**, the **OpenVLM Leaderboard** provides a toolkit to evaluate VLMs across multiple benchmarks with a single command.

What You'll Build

By the end of this module, you will:

- Learn how to use VLMs with the 🤗 transformers library
- Understand chat templates and conversation formatting for VLMs
- Fine-tune SmolVLM on your own dataset
- Run both programmatic and CLI-based training workflows

Let's dive into the fascinating world of Vision Language Models!

References

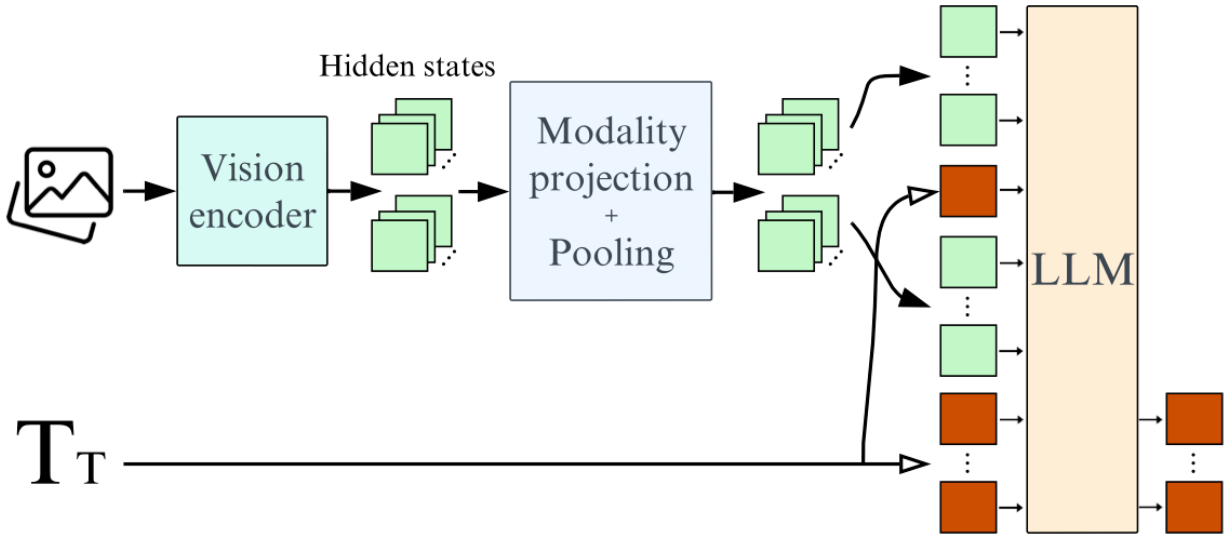
- Vision Language Models (Better, Faster, Stronger)
- Fine-Tuning a Vision Language Model (Qwen2-VL-7B) with the Hugging Face Ecosystem (TRL)
- Fine-tuning SmolVLM with TRL on a consumer GPU
- Fine-tuning SmolVLM using direct preference optimization (DPO) with TRL on a consumer GPU
- Fine tuning a VLM for Object Detection Grounding using TRL

- Fine-Tuning a Vision Language Model with TRL using MPO
- Post training a VLM for reasoning with GRPO using TRL
- Preference Optimization for Vision Language Models with TRL
- Vision Language Models Explained
- SmolVLM - small yet mighty Vision Language Model
- SmolVLM2-2.2B-Instruct
- CLIP: Learning Transferable Visual Models from Natural Language Supervision
- Align Before Fuse: Vision and Language Representation Learning with Momentum Distillation
- Object Understanding with Vision Language Models Space

Using Pretrained VLMs

Visual Language Models (VLMs) process **images and text simultaneously**, enabling advanced tasks like generating captions, answering visual questions, or reasoning across modalities. In this section, we focus on **how VLMs work and how to use them practically**.

Architecture Overview



VLMs combine image-processing and text-generation components for a unified multimodal understanding. The main elements are:

- **Image/Vision Encoder**: Converts images into compact numerical representations. Examples: CLIP, SigLIP.
- **Embedding Projector**: Aligns image features with text embeddings (often a small MLP or linear layer fine-tuned for the multimodal task).
- **Multimodal Projector / Fusion Module**: Fuses and enhances connections between visual and textual representations. This step goes beyond alignment, enabling rich cross-modal interaction.representations.
- **Text Decoder**: Generates text (or other outputs) from the fused multimodal representations.

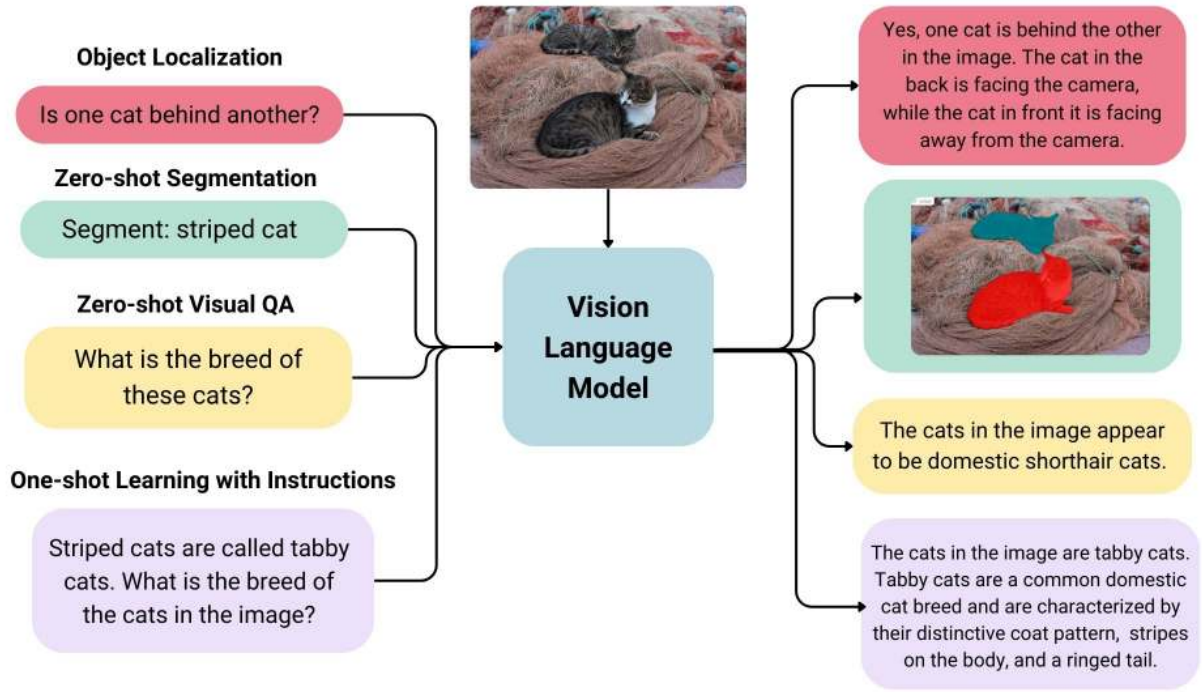
Most VLMs leverage **pretrained image encoders and text decoders**, then fine-tune on paired image-text datasets for efficient training and generalization.

Practical Usage

VLMs can be applied to tasks such as:

- **Image Captioning:** generating descriptions for images
- **Visual Question Answering (VQA):** answering questions about an image
- **Cross-Modal Retrieval:** matching images with text and vice versa
- **Creative Applications:** design, art generation, multimedia content

High-quality paired datasets are key, and 🤗 transformers provide pretrained models and streamlined fine-tuning workflows.



Chat Format

Many VLMs support **chat-like interactions**, with messages structured as:

1. **System message:** sets context: `"You are an assistant analyzing visual data."`
2. **User queries:** combine text and images.
3. **Assistant responses:** generated text based on multimodal analysis.

Example:

```
[
  ]
  },
  ,

    ]
  },
  ]
  }
]
```

VLMs can also handle **multiple images or video frames** as input by passing sequences of images through the same chat template.

Using a VLM via pipeline

As we saw in Unit 1, the easiest way to use a VLM is through the 🤗 `pipeline` abstraction:

```
from transformers import pipeline

# Initialize the pipeline with a VLM
pipe = pipeline("image-text-to-text", "HuggingFaceTB/SmolVLM2-2.2B-Instruct",
device_map="auto")

# Define your conversation with an image
messages = [
     ,
             ,
         ],
     }
 ]

outputs = pipe(text=messages, max_new_tokens=60, return_full_text=False)

# Generate response - pipeline handles multimodal inputs automatically
response = pipe(messages, max_new_tokens=128, temperature=0.7)

print(response[0]['generated_text'][-1]['content'])  # Print the model's
description
```

Output

```
The image depicts a close-up view of a flower garden, specifically focusing on a
pink flower. The flower is the central subject of the image, and it is a prominent
feature due to its vibrant color and intricate details. The flower has a circular
shape, with petals that are slightly curled and have a gradient from light to dark
pink. The petals are arranged symmetrically around the central pistil, which is
visible in the center of the flower. The pistil is a small, yellow structure that
is surrounded by a cluster of stamens, which are visible as small, yellow
structures. The flower also has a small, black
```

Using a VLM via Transformers (Full Control)

For advanced use, you can access a VLM directly via 🤗 Transformers, giving you **full control over each component**.
To reduce memory usage and speed up inference, we can apply **4-bit quantization** using `bitsandbytes`.

Unlike standard LLM usage, VLMs require a **processor** instead of just a tokenizer. The processor handles both **text tokenization** and **image preprocessing**, streamlining the workflow for multimodal inputs.

```
from transformers import AutoProcessor, AutoModelForImageTextToText,
BitsAndBytesConfig
from transformers.image_utils import load_image

device = "cuda" if torch.cuda.is_available() else "cpu"

# Quantization for efficiency
quant_config = BitsAndBytesConfig(load_in_4bit=True)
model_name = "HuggingFaceTB/SmolVLM2-2.2B-Instruct"
model = AutoModelForImageTextToText.from_pretrained(model_name,
quantization_config=quant_config).to(device)
processor = AutoProcessor.from_pretrained(model_name)
```

Example: Describe an Image

We can use the **chat template** to describe images. Each image is represented as `` `in the message, and the actual image data is passed to the processor via the` ``images` argument. The processor handles both text and visual inputs seamlessly.

```
# Load image
image_url = "https://huggingface.co/datasets/huggingface/documentation-images/
resolve/main/bee.jpg"
image = load_image(image_url)

# Create input messages
messages = [
    ,

        ]
    },
]

# Prepare inputs
prompt = processor.apply_chat_template(messages, add_generation_prompt=True)
inputs = processor(text=prompt, images=[image], return_tensors="pt")
inputs = inputs.to(device)

# Generate outputs
generated_ids = model.generate(**inputs, max_new_tokens=500)
generated_texts = processor.batch_decode(
    generated_ids,
    skip_special_tokens=True,
)[0]

# Extract only the assistant response
assistant_response = generated_texts.split("Assistant:")[-1].strip()
```

```
print(assistant_response)
```

Output

```
The image is of a bee on a flower.
```

The processor combines the text and image inputs, so the model can generate coherent multimodal outputs.

> [!TIP] Similar templates can handle multiple images, OCR tasks, or even video frames, making VLMs highly versatile.

Resources

- Vision Language Models (Better, Faster, Stronger)
- Vision Language Models Explained
- SmolVLM - small yet mighty Vision Language Model

Fine-Tuning VLMs

In Unit 1, we explored supervised fine-tuning on LLMs, including efficient strategies using TRL. In this section, we adapt these techniques for **Vision Language Models (VLMs)**, focusing on efficiency and task-specific performance.

Key Efficiency Techniques

When fine-tuning VLMs, memory and computation can quickly become a bottleneck. Here are the main strategies:

Quantization

Quantization reduces the precision of model weights and activations, lowering memory usage and speeding up computation. - **bfloat16 / float16** halves memory requirements while maintaining accuracy. - **8-bit / 4-bit quantization** reduces memory further, with minor performance trade-offs.

> ⚠️ Especially relevant for VLMs, where image features increase memory demands.

PEFT & LoRA

Low-Rank Adaptation (LoRA) freezes the base model weights and trains **compact rank-decomposition matrices**, drastically reducing the number of trainable parameters. When combined with **PEFT**, fine-tuning requires **millions of trainable parameters instead of billions**, making large VLMs accessible on limited hardware.

Batch Size Optimization

Memory-efficient training can be achieved with: - **Gradient accumulation**: maintain effective batch size over multiple steps.
- **Gradient checkpointing**: recompute intermediate activations to save memory.
- Start with a large batch, reduce if OOM errors occur, and combine with LoRA/ quantization for best results.

Supervised Fine-Tuning (SFT)

SFT adapts a pre-trained VLM to a **specific task** using labeled datasets (image-text pairs). Examples include: - Visual question answering (VQA)
- Image captioning
- Chart or diagram interpretation

When to Use SFT

- Specialize a VLM in a domain where the base model struggles.
- Learn domain-specific vocabulary or visual patterns.

Limitations

- Requires **high-quality, labeled datasets**.
- Can be **computationally intensive**.
- Risk of **overfitting** if fine-tuning is too narrow.

Usage Example

The `SFTTrainer` supports training VLMs directly.
Your dataset should include an additional `images` column containing the visual inputs. See the dataset format docs for details.

```
from trl import SFTTrainer

training_args = SFTTrainer(
   output_dir="./fine_tuned_model",
   per_device_train_batch_size=4,
   num_train_epochs=3,
   learning_rate=5e-5,
   save_steps=1000,
   bf16=True,
   gradient_checkpointing=True,
   gradient_accumulation_steps=16,
   logging_steps=50
)
```

⚠️ **Important:** Set `max_length=None` in the `SFTConfig`.
Otherwise, truncation may remove image tokens during training.

```
SFTConfig(max_length=None, ...)
```

Practical Steps

1. **Data Preparation**
2. Use image-text pairs, e.g., `HuggingFaceM4/ChartQA`.
3. **Model Setup**
4. Load a pre-trained VLM such as `HuggingFaceTB/SmolVLM2-2.2B-Instruct`.
5. Initialize a processor to prepare text and image inputs.
6. **Fine-Tuning Process**
7. Format data into **chat-like messages** (`system`, `user`, `assistant`).
8. Configure optimizer, batch size, and gradient accumulation.
9. Apply **quantization** and **LoRA** for memory-efficient training.

Preference Optimization (DPO)

Direct Preference Optimization (DPO) aligns a VLM with **human preferences** instead of strict instruction following.
- Useful for creative tasks, subjective judgments, or multi-choice answers.
- The model learns to select the **more human-aligned response**, even if it isn't strictly "correct."

Limitations

- Requires **high-quality preference-labeled datasets**.
- Training involves **pairwise preference sampling** and careful resource management.

Usage Example

- **Dataset**: Each example contains a prompt (image + question) and **two candidate responses**:

```
Question: How many families?
Rejected: The image does not provide information about families.
Chosen: The image shows a Union Organization table setup with 18,000 families.
```

- **Model Setup**: Load the pre-trained VLM, integrate with TRL DPO, and prepare the processor.
- **Training Pipeline**:
- Format dataset into chat-like messages.
- Apply a **preference-based loss function**.
- Use gradient accumulation, checkpointing, LoRA, and quantization for efficiency.

SFT vs DPO Comparison

Feature	SFT	DPO
Input	Labeled image-text	Image-text + preference-ranked outputs
Loss	Standard supervised	Preference-based
Goal	Task-specific adaptation	Human-aligned output
Use Case	Domain specialization	Creative, subjective, or multi-choice tasks

Practical Tips

- Start small: test with a **subset of the dataset** before full training.
- Use **gradient checkpointing + LoRA + quantization** to reduce memory usage.
- Monitor **checkpoint frequency** to balance storage and safety.
- Validate on a small set to **avoid overfitting**.

Next Steps

After fine-tuning, evaluate your VLM's performance on **multimodal tasks** using benchmarks and custom test sets, applying techniques from Unit 2.

Fine-Tuning a VLM in `hf jobs` using TRL

As introduced in earlier units, Hugging Face Jobs make fine-tuning Vision Language Models (VLMs) straightforward. You can run **Supervised Fine-Tuning (SFT)** or **Direct Preference Optimization (DPO)** directly on the Hugging Face infrastructure with minimal setup, adjusting the training parameters we discussed previously.

Quick Example

```
hf jobs uv run \
   --flavor a100-large \
   --secrets HF_TOKEN \
   --timeout 2h \
   "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/sft.py" \
   --model_name_or_path HuggingFaceTB/SmolVLM2-2.2B-Instruct \
   --dataset_name HuggingFaceM4/ChartQA \
   --report_to trackio
```

- `--flavor a100-large`: GPU type for training.
- `--secrets HF_TOKEN`: Your Hugging Face token.

The script handles processor setup, data formatting, and model training automatically. Once the job finishes, your fine-tuned VLM is ready to download and use in downstream tasks.

> [!TIP] For memory-efficient fine-tuning of large VLMs, consider combining techniques like **LoRA adapters**, **gradient accumulation**, and **quantization**. These strategies help reduce memory usage while maintaining performance.

Resources

- Fine-Tuning a Vision Language Model (Qwen2-VL-7B) with the Hugging Face Ecosystem (TRL)
- Fine-tuning SmolVLM with TRL on a consumer GPU
- Fine-tuning SmolVLM using direct preference optimization (DPO) with TRL on a consumer GPU
- Preference Optimization for Vision Language Models with TRL

Hands-On Exercises: Fine-Tuning SmolVLM2-2.2B-Instruct

Welcome to the practical section! Here you'll put into practice everything you've learned about vision language models (VLMs) using **HuggingFaceTB/SmolVLM2-2.2B-Instruct**.
The exercises progress from foundational concepts to advanced techniques, helping you gain real-world, hands-on experience.

Learning Objectives

By the end of these exercises, you will be able to:

- **Work with VLM datasets**: Explore and prepare **HuggingFaceM4/ChartQA**.
- **Optimize training**: Apply **quantization** and **PEFT** for efficient fine-tuning.
- **Fine-tune models in practice**: Train **HuggingFaceTB/SmolVLM2-2.2B-Instruct** using both Python APIs and CLI tools.
- **Adapt datasets for TRL**: Prepare VLM datasets to integrate seamlessly with TRL workflows.
- **Move to production**: Understand how to scale and manage **production-ready fine-tuning workflows** for VLMs.

Exercise 1: Explore `SmolVLM2-2.2B-Instruct`

Objective: Get familiar with the `SmolVLM2-2.2B-Instruct` model and evaluate the model using a sample from the dataset.

Environment Setup

> [!WARNING] - You need a GPU with at least 8GB VRAM for training. CPU/MPS can run formatting and dataset exploration, but training larger models will likely fail. - First run will download several GB of model weights; ensure 15GB+ free disk and a stable connection. - If you need access to private repos, authenticate with Hugging Face Hub via `login()`.

First, install the required libraries: `transformers`, `datasets`, `trl`, `huggingface_hub`, and `trackio`.
These packages provide the tools for working with the model, datasets, and Hugging Face Hub.

```
# Install required packages (run in Colab or your environment)
pip install transformers datasets trl huggingface_hub trackio num2words==0.5.14
```

Import dependencies

Now, import the main dependencies we'll use:

```
# Import dependencies



from transformers import AutoProcessor, AutoModelForImageTextToText,
BitsAndBytesConfig
from transformers.image_utils import load_image
```

Load the model and processor

1. Select the device

We start by selecting the device where the model will run. It can be a GPU (`cuda`), Apple Silicon (`mps`), or the CPU as a fallback.

```
device = (
    "cuda"
```

```
    if torch.cuda.is_available()
    else "mps" if torch.backends.mps.is_available() else "cpu"
)
```

2. Authenticate with Hugging Face

To work with private models or to **push your fine-tuned model to the Hub** (as we'll do in this exercise), you need to authenticate with your Hugging Face account.

> [!TIP] You can create and copy your access token from the Hugging Face tokens page in your profile.

```
from huggingface_hub import login
login()
```

3. Load the model and processor

Finally, we load the **HuggingFaceTB/SmolVLM2-2.2B-Instruct** model. The `AutoProcessor` is also initialized here — it ensures that both text and images are preprocessed correctly before being passed to the model.

```
model_name = "HuggingFaceTB/SmolVLM2-2.2B-Instruct"
model = AutoModelForImageTextToText.from_pretrained(
    model_name,
    dtype=torch.bfloat16,
).to(device)

processor = AutoProcessor.from_pretrained(model_name)
```

Explore the dataset

In this step, we load a small subset of the **ChartQA** dataset — just 10% of the training and validation splits — to keep the exercises fast and manageable for learning purposes.

We then display one of the chart images using `matplotlib` to get a visual sense of the model's input.
Additionally, we print the corresponding query and label so you can fully understand the dataset structure and the type of tasks the model will handle.

```
from datasets import load_dataset

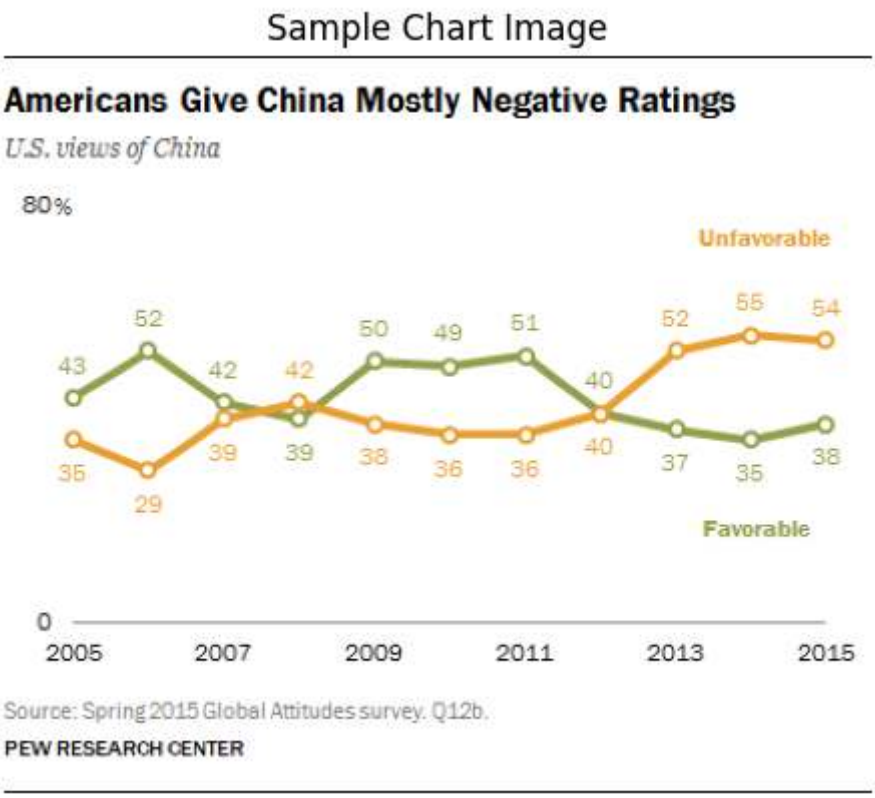

train_dataset, eval_dataset = load_dataset("HuggingFaceM4/ChartQA", split=["train[:
10%]", "val[:10%]"])
example = train_dataset[1]
image = load_image(example["image"])

print(example["query"])
print(example["label"][0])
```

Output

```
How many values are below 40 in Unfavorable graph?
6
```

```
plt.imshow(image)
plt.axis("off")
plt.title("Sample Chart Image")
plt.show()
```



Build a chat-style prompt

We create a **chat message list** that includes a user query along with the image.

Using `processor.apply_chat_template`, we transform this into the exact input format the model expects.

```
# Define a chat-style prompt
messages = [
    ,
        ,
    ]}
]

# Apply the chat template
chat_prompt = processor.apply_chat_template(
      messages, add_generation_prompt=True\
)
print(chat_prompt)
```

Output

```
<|im_start|>User:<image>How many values are below 40 in Unfavorable graph?
<end_of_utterance>
Assistant:
```

Run inference

We tokenize the chat prompt and image into tensors, then generate a response with the model. Finally, we decode the output tokens back into text.

```
# Tokenize input
inputs = processor(images=[image], text=chat_prompt,
return_tensors="pt").to(device)

# Generate model output
with torch.no_grad():
    output = model.generate(**inputs, max_new_tokens=20)

# Trim the generated ids to remove the input ids
trimmed_generated_ids = [out_ids[len(in_ids) :] for in_ids, out_ids in
zip(inputs.input_ids, output)]

# Decode the output text
output_text = processor.batch_decode(
    trimmed_generated_ids, skip_special_tokens=True,
clean_up_tokenization_spaces=False
)
print(output_text[0])
```

Output

```
3.
```

The model generates a reponse, but it's not exactly correct. It could be improved with some fine-tuning. Now that we've seen how to build prompts and generate responses with **SmolVLM2-2.2B-Instruct**, it's time to learn how to adapt and **fine-tune** the model efficiently using **LoRA (Low-Rank Adaptation)**. This approach allows training large models with fewer resources and prepares the model for specific downstream tasks.

Exercise 2: Fine-Tune the Model Using LoRA

In this exercise, we'll apply **LoRA (Low-Rank Adaptation)** to fine-tune our Vision-Language Model efficiently.

LoRA works by injecting **trainable low-rank matrices** into existing model layers, enabling **large models to be fine-tuned with significantly fewer trainable parameters**.
This approach reduces memory usage and speeds up training while maintaining high performance.

```
system_message = """You are a Vision Language Model specialized in interpreting
visual data from chart images.
Your task is to analyze the provided chart image and respond to queries with
concise answers, usually a single word, number, or short phrase.
The charts include a variety of types (e.g., line charts, bar charts) and contain
colors, labels, and text.
Focus on delivering accurate, succinct answers based on the visual information.
Avoid additional explanation unless absolutely necessary."""
```

We'll format the dataset into a **chatbot-style structure**, where each example includes:

- A **system message** defining the assistant's role
- The **chart image**
- The **user query**
- The **expected answer**

This is the format expected by the `SFTTrainer`, including the `images` and `messages` columns.
You can learn more about preparing datasets for VLM post-training in the documentation.

Format the Dataset

The first step is to structure the dataset for VLM training.
We'll define a **system message** that instructs the model to act as a **chart analysis expert**, providing **concise, accurate answers** about chart images.

```
def format_data(sample):
    return ],
            },
            ,
                    ,
                ],
            },
            ],
            },
        ],
    }
```

Now, let's format the data using the chatbot structure. This will set up the interactions for the model.

```
train_dataset = [format_data(sample) for sample in train_dataset]
eval_dataset = [format_data(sample) for sample in eval_dataset]
```

Configure LoRA

Here we define a `LoraConfig`: - `r` and `lora_alpha` control the rank and scaling of the adaptation matrices. - `target_modules` specifies which parts of the model to adapt. - `task_type` is set for causal language modeling.

We then apply LoRA to the base model using `get_peft_model` and print out the trainable parameters to verify the adaptation.

```
from peft import LoraConfig, get_peft_model

# Configure LoRA
peft_config = LoraConfig(
    lora_alpha=16,
    lora_dropout=0.05,
    r=8,
    target_modules=["q_proj", "v_proj"],
    task_type="CAUSAL_LM",
)

# Apply PEFT model adaptation
peft_model = get_peft_model(model, peft_config)
```

```
# Print trainable parameters
peft_model.print_trainable_parameters()
```

Set up the Trainer

We configure the **SFTTrainer** from `trl` with `SFTConfig`: - `num_train_epochs`, `batch_size`, and `gradient_accumulation_steps` control the training loop. - `gradient_checkpointing`, and `bf16` optimize memory and speed. - `learning_rate` manages optimization. - `train_dataset` and `eval_dataset` are aligned with your dataset.

This prepares the trainer to handle fine-tuning with PEFT/LoRA.

```
from trl import SFTConfig, SFTTrainer

# Configure training arguments using SFTConfig
training_args = SFTConfig(
    output_dir="smol-course-smolvlm2-2.2b-instruct-trl-sft-ChartQA",
    num_train_epochs=1,
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4,
    learning_rate=1e-4,
    logging_steps=25,
    save_strategy="steps",
    save_steps=25,
    optim="adamw_torch_fused",
    bf16=True,
    push_to_hub=True,
    report_to="trackio",
    max_length=None,
)

# Initialize the Trainer
trainer = SFTTrainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset,
    peft_config=peft_config,
)

# Align the SFTTrainer params with your chosen dataset.
```

Train and Save the Model

Now we run the training loop: 1. `trainer.train()` starts fine-tuning with LoRA. 2. `trainer.save_model()` stores the locally trained model.

This step ensures the model is ready for downstream tasks with minimal additional parameters.

```
# Train the model
trainer.train()

# Save the model
trainer.save_model(training_args.output_dir)
```

With the foundations of Python-based fine-tuning and LoRA in place, we can now move this workflow to a **production environment** using the **TRL CLI**. This approach lets you automate fine-tuning and create reproducible pipelines without writing full Python scripts.

Exercise 3: Production Workflow with TRL CLI

In the previous exercises, we focused on using the Python API to fine-tune **SmolVLM2-2.2B-Instruct**, exploring dataset preparation and generating chat-style prompts.

In this exercise, we'll demonstrate how to perform fine-tuning using the **TRL CLI**, a common workflow in production environments. The CLI allows you to run experiments and manage training without writing Python scripts. If you want a refresher, we previously introduced this tool here, and the same concepts and troubleshooting tips apply.

The TRL CLI leverages the same logic and configuration options as the Python API but presents them through a simple command-line interface. This means you can define everything—from the model and dataset to training hyperparameters and output location—in a single command.

The example below shows how to fine-tune **SmolVLM2-2.2B-Instruct** on the **trl-lib/llava-instruct-mix** dataset, using LoRA for parameter-efficient fine-tuning, mixed precision for faster training, and optional push-to-Hub for sharing your model. The dataset now is different.

We are using a different dataset here because it already comes formatted in the expected VLM structure, as discussed earlier.

- `--model_name_or_path` specifies the base model to fine-tune.
- `--dataset_name` and `--dataset_config` define the dataset and subset.
- `--output_dir` sets the local directory for saving the fine-tuned model.

- `--per_device_train_batch_size` and `--gradient_accumulation_steps` control effective batch size and memory usage.
- `--learning_rate`, `--num_train_epochs`, and `--max_length` define the core training hyperparameters.
- `--bf16` enables mixed precision for faster and more memory-efficient training on compatible GPUs.
- `--push_to_hub` and `--hub_model_id` allow automatic upload of the trained model to your Hugging Face Hub repository.

Using the TRL CLI is functionally equivalent to writing a full Python training script, but it's faster to configure, easier to reproduce, and ideal for production pipelines or automated training workflows.

```
trl sft \
    --model_name_or_path HuggingFaceTB/SmolVLM2-2.2B-Instruct \
    --dataset_name trl-lib/llava-instruct-mix \
    --output_dir ./smolvln-instruct-sft-cli \
    --per_device_train_batch_size 1 \
    --gradient_accumulation_steps 16 \
    --learning_rate 2e-4 \
    --num_train_epochs 3 \
    --max_length -1 \
    --logging_steps 5 \
    --save_steps 100 \
    --warmup_steps 50 \
    --bf16 True \
    --push_to_hub \
    --hub_model_id your-username/smolvlm2-2.2b-instruct-sft-cli
```

Exercise 4: Training with Hugging Face Jobs

In Unit 1, we introduced **Hugging Face Jobs (HF Jobs)** and demonstrated how to fine-tune a model using this managed cloud service.
HF Jobs provides a **fully managed infrastructure** for training models, eliminating the need to set up GPUs, manage dependencies, or configure environments locally. This is especially useful for **SFT training**, which can be both resource-intensive and time-consuming.

Following the same approach, we can use HF Jobs to fine-tune our Vision-Language Model (VLM).
If needed, refer back to Unit 1 to refresh your understanding of HF Jobs and their workflow.

Here’s an example of how to launch a training job using **TRL’s maintained SFT script**:

```
# Use TRL's maintained SFT script directly
hf jobs uv run \
    --flavor a10g-large \
    --timeout 2h \
    --secrets HF_TOKEN \
    --with num2words==0.5.14 \
    "https://raw.githubusercontent.com/huggingface/trl/main/trl/scripts/sft.py" \
    --model_name_or_path HuggingFaceTB/SmolVLM2-2.2B-Instruct \
    --dataset_name trl-lib/llava-instruct-mix\
    --learning_rate 5e-5 \
    --per_device_train_batch_size 4 \
    --max_length -1 \
    --max_steps 1000 \
    --output_dir smolvlm2-2.2b-instruct-sft-jobs \
    --push_to_hub \
    --hub_model_id your-username/smolvlm2-2.2b-instruct-sft-jobs \
    --report_to trackio
```

After launching the job, HF Jobs will handle the **entire training process** in the cloud. You can monitor progress, view logs, and track metrics directly from the Hugging Face Hub.

Once the job completes:

- The **fine-tuned model** will be available in the `output_dir` you specified.
- If `--push_to_hub` was used, the model will also be **accessible from your Hugging Face account**, ready for inference or further fine-tuning.
- You can **resume, replicate, or scale** training easily by re-running or modifying the job configuration.

This workflow removes the overhead of managing local resources, allowing you to focus on **model experimentation and evaluation**.

Test your knowledge

You’ve completed the unit — great work! Now put your learning to the test by taking the quiz.

Resources for Further Learning

Here are some helpful resources to deepen your understanding and continue experimenting with vision language models and TRL workflows:

- **TRL Documentation** – Complete reference for using TRL, including Python API and CLI.
- **HuggingFaceTB/SmolVLM2-2.2B-Instruct Model Card** – Detailed information about the model architecture, training, and usage.
- **HuggingFaceM4/ChartQA Dataset** – Dataset used for training and fine-tuning VLMs.
- **Hugging Face Hub** – Platform to share your fine-tuned models and discover community models.
- **Hugging Face Discord Community** – Join the community for discussions, support, and troubleshooting.